

Modern LLM Architecture

A Visual Guide to Attention, Memory, Sparsity, and Hybrid Models – Seven-Chapter Review Pilot

ML/LLM Study Workbooks

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1 How to use this workbook

Each chapter follows the same structure: learning objectives, why the topic matters, a mental model, a visual overview, the technical core (equations with every symbol defined), a fully worked numeric example, a comparison table, comprehension questions, a common misconception, an applied exercise, an interview-style question, a chapter recap, and sources. Short questions appear near the concept they test, with space left for written answers if you're using this as a workbook rather than only a reading; full solutions are collected in the answer key (Section 10, Section 11, Section 12, Section 13, Section 14, Section 15, Section 16) rather than revealed inline, so you can attempt each question before checking it. A notation quick reference (Section 2) covers every symbol Chapters 1-7 use.

This is a seven-chapter pilot, not the complete workbook

This document contains **only Chapters 1 through 7** of *Modern LLM Architecture: A Visual Guide to Attention, Memory, Sparsity, and Hybrid Models* (workbook 04 of the ML/LLM Study Workbooks series). Chapter 8 – a synthesis/practice chapter – is planned but **not drafted**. Nothing in this document should be read as implying it exists yet. Multi-head latent attention (MLA), briefly deferred out of Chapter 3’s drafting pass, is now covered in Chapter 4 alongside sliding-window and sparse attention. Chapter 4 covers recurrent/state-space/linear-attention alternatives only as a brief taxonomy and forward reference; Chapter 6 adds the primary source that taxonomy was pending (Gu and Dao’s selective state-space mechanism) once a hybrid case study needed it. Chapter 5 covers mixture-of-experts architecture only – detailed distributed-training implementation, collective-communication algorithms, kernel optimization, inference-engine configuration, expert quantization, model-specific benchmark comparisons, post-training, and routing-research surveys are all deferred to later workbooks or a future revision. Chapter 6 selects exactly three real, pinned model families (Llama 3 405B, DeepSeek-V3, Jamba) as case studies – it is not a survey of every current model, not a benchmark ranking, and not a release-history timeline; each model’s architecture facts are dated and traceable to that model’s own technical report, not restated as permanent or “latest” facts. Chapter 7 synthesizes systems consequences already implied by Chapters 2-6 (a resource ledger, prefill-vs-decode, an architecture-to-systems consequence map, and a minimal parallelism overview); it previews looped depth in one row only (full treatment deferred to Chapter 8) and cross-references multi-token prediction and speculative decoding without teaching either mechanism – both are assigned to future workbooks in this series’ own cross-workbook topic roadmap, not taught here.

This is a review artifact: it exists to establish the workbook’s explanatory depth, visual language, page density, question style, and citation style before the remaining chapter is drafted at scale. Treat any rough edge you find as exactly the kind of thing this pilot is meant to surface early.

2 Notation quick reference

This project maintains a full, growing symbol table across all eight workbooks; later chapters introduce mechanisms this pilot doesn’t cover. The subset below is everything Chapters 1-5 actually use – flip back here if a symbol in an equation isn’t immediately obvious.

3 Chapter 1: Transformer Refresher

From Tokens to the Next-Token Distribution

3.1 Learning objectives

By the end of this chapter you should be able to:

- State the decoder-only next-token objective and explain why it is what lets a model generate text one piece at a time.
- Trace a single token’s vector as it moves through the residual stream, the attention sublayer, and the feed-forward sublayer of one decoder block.
- Explain causal masking precisely – which positions a given position may and may not attend to – rather than the imprecise “only the previous token” version.
- Distinguish prefill from a single autoregressive decode step, and explain concretely what state gets reused between decode steps.

3.2 Why this matters

Every mechanism in the rest of this workbook – attention variants, positional schemes, sparsity, mixture-of-experts routing – is a modification to the same handful of moving parts introduced here: a residual stream, an attention sublayer, a feed-forward sublayer, and a causal mask. If any of those four feel shaky, later chapters will read as a list of unrelated tricks rather than variations on one theme. This chapter’s job is narrow on purpose: get the shared vocabulary and the prefill/decode distinction solid, so that “GQA changes how many KV heads exist” (Chapter 3) or “MoE changes which parameters activate” (Chapter 5) land as small, precise edits to a picture you already have, not new pictures from scratch.

3.3 Mental model or analogy

Think of the residual stream as a running total printed on a receipt. Each sublayer in the model doesn’t erase what’s already printed – it adds one more line of adjustment underneath the current total. By the time the receipt reaches the bottom of the block, the final total already reflects every earlier adjustment, because each one was added in, in order.

Analogy

The residual stream is a running total on a receipt. Each sublayer reads the current total, computes an adjustment, and prints a new line that *adds* to the total – it never crosses out or overwrites a previous line. Where this breaks down: a paper receipt keeps each line individually readable, so you can always look back and see exactly what line 3 contributed. The residual stream cannot do that. Every adjustment is summed into the *same* fixed-width vector, so by the end you have one number (well, one vector) that reflects every contribution – but you cannot read back which sublayer contributed what just by looking at the final total. That specific limitation is the working assumption behind why interpretability researchers need extra tools (activation patching, probing) to recover what an individual sublayer contributed – the sum alone doesn’t show its work.

3.4 Visual overview

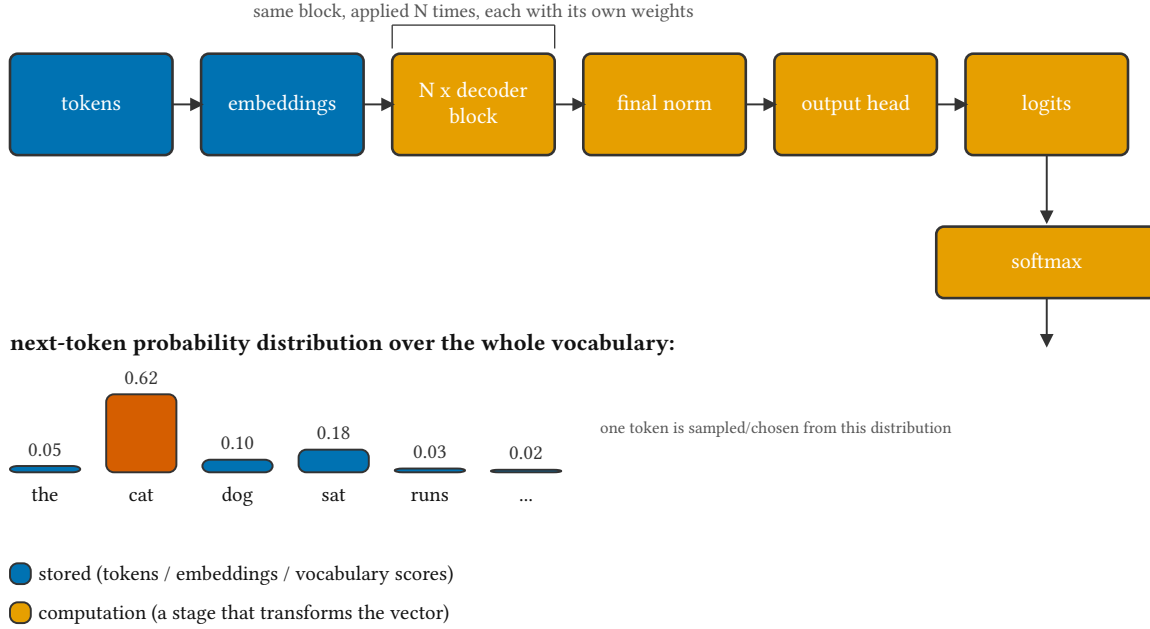


Figure 1: Tokens become vectors, the same block repeats N times with its own weights at each repetition, and only the final softmax turns a vector back into a probability distribution over the vocabulary – nothing “decides” the next token before that last step.

3.5 Technical core

Tokens and embeddings. A tokenizer splits text into a sequence of S discrete tokens, each an integer id into a fixed vocabulary of size V . An embedding table $E \in \mathbb{R}^{V \times d_{\text{model}}}$ maps each token id to a dense vector of width d_{model} (the **hidden dimension**, the width of the residual stream everywhere in the model). For a batch of B sequences of length S , the embedded input has shape (B, S, d_{model}) . We will mostly drop the batch dimension B below and reason about one sequence at a time – everything below happens identically and independently across the batch dimension.

$$x_i = E[\text{token_id}_i], \quad x_i \in \mathbb{R}^{d_{\text{model}}} \quad (1)$$

In words: look up a row of a table. That row is where the model’s only “knowledge” of which token this is enters the computation – from here on, everything operates on vectors, not token identities.

The residual stream. Stack the S embedded vectors into $x \in \mathbb{R}^{S \times d_{\text{model}}}$. This is the residual stream at the input to block 1. Every block updates it the same way:

$$x^{(l+1)} = x^{(l)} + f(\text{Norm}(x^{(l)})) \quad (2)$$

where f is either the attention sublayer or the feed-forward sublayer, and Norm is a normalization step (Chapter 2 covers exactly which normalization and exactly where it sits; for now, treat it as “rescale the vector before reading it”). *In words:* a sublayer never overwrites $x^{(l)}$ – it reads a normalized copy, computes an update, and the update is *added back* onto the original, un-normalized $x^{(l)}$. Shape-wise, $f(\cdot)$ must return something of shape (S, d_{model}) , the same shape it was handed, or the addition would not type-check. That shape constraint is exactly why d_{model} stays constant across every block in a model.

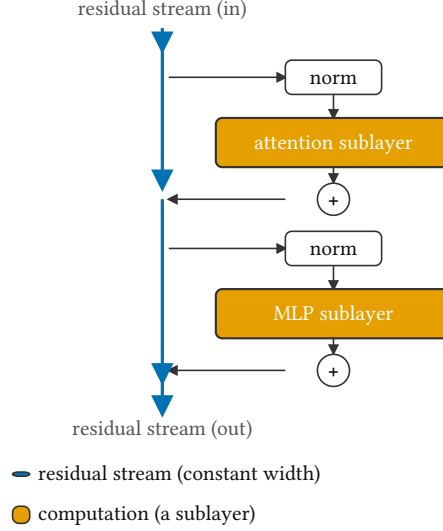


Figure 2: One block, showing the residual stream as the vertical bar and the two sublayers branching off it. Notice the stream’s width never changes – only what gets added to it does.

The attention sublayer. Each block’s attention sublayer projects the normalized stream into three role-specific views:

$$Q = \tilde{x}W_Q, \quad K = \tilde{x}W_K, \quad V = \tilde{x}W_V \quad (3)$$

where $\tilde{x} = \text{Norm}(x)$, and $W_Q, W_K, W_V \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ are learned weight matrices. Q, K, V each have shape (S, d_{model}) . Rather than using the full width at once, attention splits d_{model} into H_q **heads** of width $d_{\text{head}} = d_{\text{model}}/H_q$, so each of Q, K, V is reshaped to $(H_q, S, d_{\text{head}})$ and every head runs the attention computation below independently, in parallel, on its own d_{head} -wide slice. *In words:* Q asks “what is this position looking for,” K answers “what does this position offer,” and V is “what this position actually contributes if attended to” – three different linear views of the same vector, learned separately.

Notation note

We write H_q , not just H , for the number of heads – specifically the number of *query* heads. In this chapter every head has its own matching key/value head, so there is no need yet to distinguish a separate key/value head count. Chapter 3 introduces designs (grouped- query and multi-query attention) where the query head count and the key/value head count are genuinely different numbers, at which point the “q” in H_q starts pulling its weight notationally.

For one head, scaled dot-product attention is:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_{\text{head}}}} + M \right) V \quad (4)$$

$QK^\top$ has shape (S, S) – one raw compatibility score per query/key pair. Dividing by $\sqrt{d_{\text{head}}}$ keeps those scores from growing too large as d_{head} grows (large scores would push softmax toward a near-one-hot output before the model has learned anything useful). M is the **causal mask**, and it is the entire reason a decoder can be trained on whole sequences at once without cheating:

$$M_{ij} = \begin{cases} 0 & \text{if } j \leq i \\ -\infty & \text{if } j > i \end{cases} \quad (5)$$

In words: position i 's query is allowed to see keys from position i itself and every position before it ($j \leq i$); a key from any later position ($j > i$) gets an additive $-\infty$, which softmax then sends to exactly zero weight. This is not “attend only to the previous token” – it is “attend to every earlier token, plus yourself, and nothing later.” Figure 3 below makes the difference concrete: notice that the shape of the allowed region is a solid triangle, not a single diagonal line.

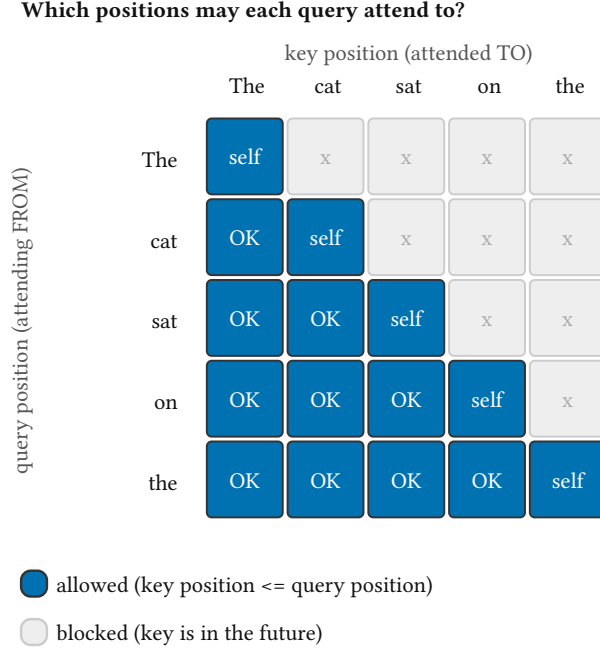


Figure 3: Causal visibility for the chapter’s running example: the allowed region (blue) is everything at or before a row’s own position, not a single step back – position 5 (“the”) can see all 4 prompt tokens plus itself.

After softmax, each row of the (S, S) weight matrix sums to 1 and is used to take a weighted average of V ’s rows, producing a (S, d_{head}) output per head. The H_q heads’ outputs are concatenated back to (S, d_{model}) and passed through one more learned projection W_O before being added back onto the residual stream, per Equation 2.

The feed-forward sublayer. The other sublayer, f in Equation 2, is a position-wise feed-forward network: a small two-matrix (or, as Chapter 2 will show, sometimes three-matrix) network applied independently to every position’s vector, with no cross-position mixing at all – all the cross-position mixing in a block happens in the attention sublayer.

Logits and next-token prediction. After the last block, the residual stream at the final position, $x_S^{(L)} \in \mathbb{R}^{d_{\text{model}}}$, is projected by an output head $W_{\text{out}} \in \mathbb{R}^{d_{\text{model}} \times V}$ into a vector of **logits** over the vocabulary, then turned into a probability distribution:

$$p(\text{next token} = v) = \text{softmax}(x_S^{(L)} W_{\text{out}})_v \quad (6)$$

In words: one more linear projection turns “the model’s final understanding of this position” into one raw score per vocabulary entry, and softmax turns those scores into a valid probability distribution the model (or a sampling procedure) can pick a token from. Nothing about this step is special to the *last* position – every position produces its own next-token distribution, but only the last position’s distribution is the one used to actually extend the sequence, since the earlier positions’ “next tokens” are already fixed by the real input.

Working approximation

This chapter presents one decoder block’s attention and feed-forward sublayers as the complete story. A real model stacks L of these blocks, each with its own weights, and Chapter 2 will show that the normalization step glossed over here (**Norm** in Equation 2) has a specific, consequential form and placement. Treat everything in this chapter as the shared skeleton that later chapters attach real components to – it is correct as far as it goes, but it goes only as far as “here is the skeleton every modern decoder shares.”

3.6 Worked example

We trace a complete, tiny causal decoder over the prompt “The cat sat on” (4 tokens, processed by prefill) followed by one decode step that generates a 5th token, “the”. Every number below is verified programmatically, not hand-derived – see the Build and Version Note for how.

Toy configuration (deliberately tiny; a real model’s dimensions are hundreds to thousands of times larger, but the mechanics are identical): $d_{\text{model}} = 4$, $H_q = 2$ heads of $d_{\text{head}} = 2$ each, one block. Toy (untrained, hand-picked) embeddings:

Token	Embedding
The	(1, 0, 1, 0)
cat	(0, 1, 0, 1)
sat	(1, 1, 0, 0)
on	(0, 0, 1, 1)
the	(1, 0, 0, 1)

Before tracing the numbers, the same distinction shown here holds at any scale: prefill happens once (conceptually), decode steps happen once per generated token, and each decode step only ever adds one new position’s worth of state.

Implementation-dependent detail

Conceptually, prefill evaluates the prompt positions together and constructs their per-layer K/V state, which is the model this chapter uses throughout. A production serving engine may split a long prompt into chunks for scheduling or memory reasons – chunking does not change the logical distinction this chapter draws between prompt processing and one-position-at-a-time decoding; it is an engineering detail about how “together” gets scheduled, not a different mechanism.

Prefill: build the whole KV-cache from the prompt in one pass

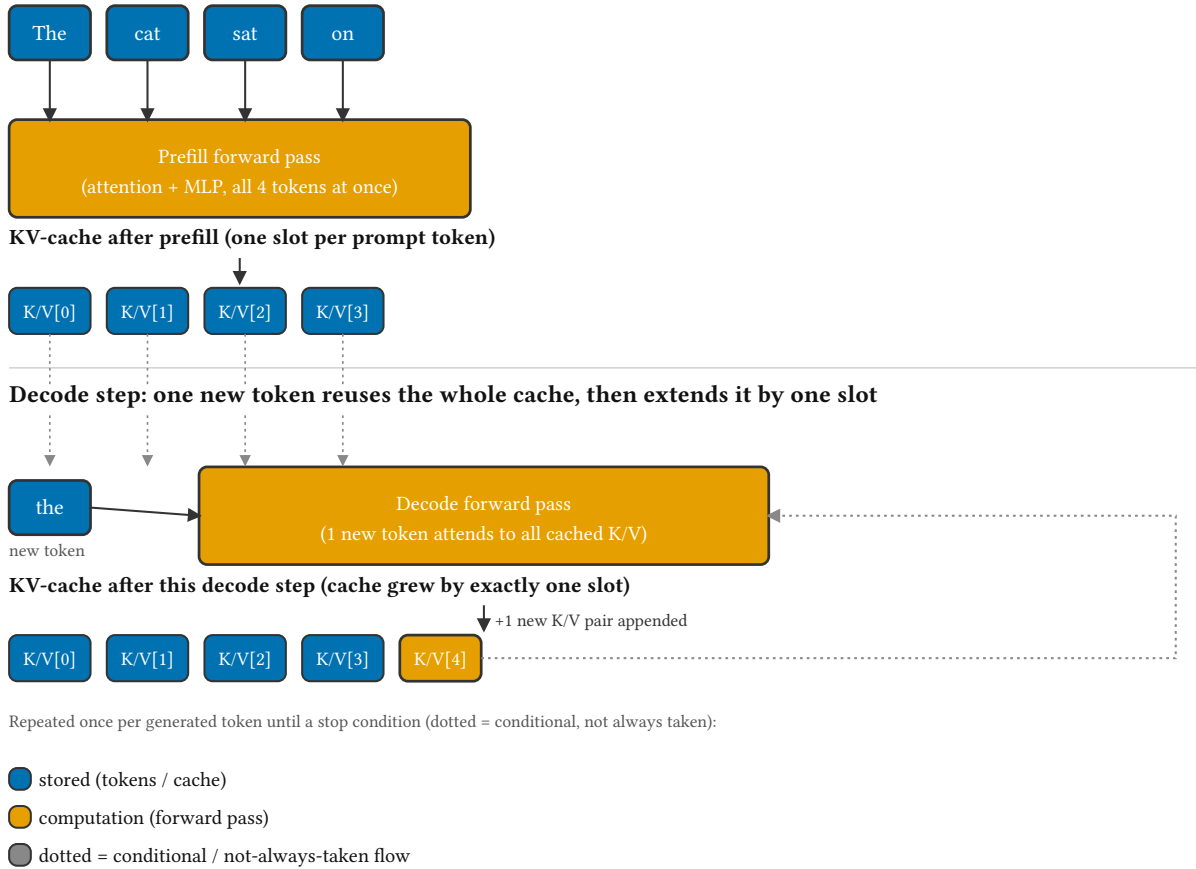


Figure 4: Prefill builds the whole cache from the prompt in one pass; each decode step afterward reads the entire cache but appends exactly one new K/V slot to it. This figure is reused as-is from the project’s existing KV-cache demonstration – the mechanics don’t change between this chapter’s introduction of the idea and Chapter 3’s deeper treatment of it.

Prefill. All 4 prompt tokens are processed together. After computing Q , K , V for all 4 positions and applying the causal mask, head 0’s softmax attention weights are:

Query position	attends to	The	cat	sat	on
The (pos 1)	1.000	–	–	–	–
cat (pos 2)	0.500		0.500	–	–
sat (pos 3)	0.140		0.576	0.284	–
on (pos 4)	0.109		0.449	0.221	0.221

Every row sums to 1.0, and every blocked (future) position is correctly absent, not just small – Equation 5 sent those entries to $-\infty$ before the softmax ever ran. Notice **sat** (row 3) puts most of its weight (0.576) on **cat**, not on itself or on **The**; that specific number is a direct, checkable consequence of the toy W_Q/W_K matrices, not a property of causal masking itself – causal masking only decides *which* positions are eligible, never how the weight is split among them.

Prefill’s output for this block includes the full (4, 4) residual stream after the attention sublayer’s update has been added in, but that residual stream is *not* what gets cached. What actually persists for reuse is each layer’s own K and V projections – computed from that layer’s input, before the attention sublayer’s output is folded back into the residual stream (Chapter 3 gives this retained state a name: the KV-cache). The residual stream is recomputed fresh at every step from whichever tokens are currently being processed; only the per-layer K/V rows persist unchanged across steps. (Some architectures apply a position-dependent transform to K before it is cached – Chapter 3 covers this – so the cached row is not always identical to the raw projection shown here, but it is still derived once per position and reused, never re-derived from the residual stream.)

Decode step. To generate token 5, the model embeds only the new token `the`, computes *its own* Q, K, V (a single new row, not a (5, 4) recomputation), and reuses the 4 cached K/V rows from prefill unchanged. Head 0’s attention weights for this one new query, over all 5 now-available keys, are:

Query position	The	cat	sat	on	the (self)
the (pos 5)	0.075	0.310	0.153	0.153	0.310

Two things to check against Equation 5: this row has *no* blocked entries, because position 5 is the last position and every other position is at or before it; and the weights still sum to 1.0 across all 5 entries. The state reused from prefill (K/V for positions 1 through 4) is bit-for-bit identical to what prefill produced – generating token 5 did not touch positions 1 through 4 at all.

3.7 Compare and contrast

Table 2: Encoder-decoder vs. decoder-only: this workbook’s scope is decoder-only, stated explicitly rather than left implicit

Aspect	Encoder-decoder	Decoder-only
Objective	An encoder reads the full input bidirectionally; a separate decoder generates output conditioned on it	One stack reads and generates using the same causally-masked self-attention throughout
Typical use	Translation, summarization with a distinct source/target	Open-ended text generation, chat, most current general-purpose LLMs
Covered in this workbook	Not covered – named here only for contrast	Every remaining chapter assumes this architecture

3.8 Check your understanding

Full solutions are in the answer key (Section 10), not below, so you can attempt these before checking:

- Quick check.** What does the residual stream carry from one block to the next – the raw output of the most recent sublayer, or something else? _____
- Explain.** Why does a decoder-only objective let the model generate text one token at a time, in a way that a bidirectional encoder’s objective does not? _____

3. **Calculation.** For a toy config with $d_{\text{model}} = 8$, $H_q = 2$ heads, and $S = 5$: what is d_{head} , and what is the shape of the raw attention score matrix $QK^\top$ for one head, before masking?
4. **Misconception check.** True or false, and why: “causal masking means each token can only attend to the immediately previous token.”
5. **Misconception check.** True or false, and why: “the residual stream is a separate memory database the model writes facts into and reads back later.”

(Space intentionally left for written answers if you are using this as a workbook exercise, not only a reading.)

Common misconception

“The Transformer generates the entire response in one forward pass.” It does not. One forward pass produces exactly one next-token distribution, for one additional position. A 50-token response requires at least 50 forward passes (one prefill pass over the prompt, then one decode step per generated token) – the model has no mechanism for “seeing ahead” to token 50 while generating token 2. This misconception is worth taking seriously because it is exactly backward from how latency actually behaves in production: generation cost scales with *response* length, almost independently of how long the *prompt* was processed in a single prefill pass.

3.9 Applied exercise

Pick a toy config of your own: choose d_{model} , number of layers L , and number of heads H_q (make sure d_{model} is divisible by H_q). For a single token’s forward path through **one** full block, write down the shape of every intermediate tensor: the input vector, $Q/K/V$ before and after splitting into heads, the per-head attention output, the concatenated output, and the vector added back onto the residual stream. Check your shapes against Equation 3 and Equation 4 – every shape should be traceable to d_{model} and H_q alone. If you want a check with real numbers instead of just shapes, redo the trace above with your own d_{model}/H_q and toy embeddings; the same programmatic verification approach applies.

3.10 Interview lens

A common interview framing is: “Walk me through what happens differently between processing a 500-token prompt for the first time versus generating the 501st token.” A strong answer distinguishes prefill (conceptually, all 500 positions evaluated together, building per-layer K/V state for the whole prompt – possibly split into chunks by the serving engine, which doesn’t change the logical picture) from a decode step (one new position, reusing the 500 cached K/V rows unchanged, only computing K/V for the new token) – and explicitly rejects “the model re-reads the whole prompt every time,” which is the same misconception as this chapter’s headline one, just phrased in systems terms. See workbook 02 (ML Interview Practice) for more on how to structure this kind of answer under time pressure.

3.11 Chapter recap

- A decoder-only model conditions strictly on earlier positions; nothing about its objective lets it “see ahead.”
- The residual stream is additive and constant-width: sublayers add updates to it, they never replace it or change its dimensionality.
- Attention has three roles (Q , K , V), a causal mask that allows every earlier-or-equal position (not just the immediately previous one), and a softmax that turns raw scores into a weighted average.

- Prefill conceptually evaluates the whole prompt together and produces cached per-layer K/V state (the residual stream itself is not that cached state); each decode step reuses that cached state and adds exactly one new position's worth of new state.
- Generating N tokens costs at least N forward passes – there is no single-pass shortcut.

3.12 Sources and further reading

Primary source for this chapter: [1], the original Transformer paper, which this chapter's notation and equations follow directly. Secondary/supporting: [2], used here only for the encoder-decoder vs. decoder-only framing convention, not for any equation.

4 Chapter 2: Anatomy of a Modern Decoder

The Components Behind Today's LLMs

4.1 Learning objectives

By the end of this chapter you should be able to:

- Explain RMSNorm's formula and state precisely what it drops relative to LayerNorm.
- Explain RoPE's rotation well enough to say why it encodes *relative* rather than absolute position.
- Explain why a gated feed-forward layer needs three weight matrices instead of two, and what that costs in parameters.
- For any one of RMSNorm, pre-norm placement, RoPE, SwiGLU, or tied embeddings, say whether it is an architectural convention (a common but non-forced choice) or something closer to a fundamental requirement – and justify the answer rather than asserting it.

4.2 Why this matters

Chapter 1 gave you the skeleton every decoder shares: a residual stream, an attention sublayer, a feed-forward sublayer, a causal mask. If you opened a 2017 Transformer and a 2025-era open-weight model side by side, the skeleton would look identical – and almost every detail *inside* it would be different. This chapter fills in what actually changed, component by component, and – just as importantly – asks of each one whether it's load-bearing or just a common convention. That distinction matters practically: a convention is something you could swap out without breaking the model's ability to function; a requirement is something the architecture depends on. Confusing the two is a fast way to either over-explain a trivial implementation choice or under-explain something that quietly determines a model's memory footprint.

4.3 Mental model or analogy

This chapter reuses Chapter 1's residual-stream picture rather than introducing a second, competing analogy – the “running total on a receipt” from Chapter 1 still describes exactly what's happening here. What's new in this chapter is *which specific operation* fills each slot in that picture (Figure 5 below), not a new mental model for the stream itself.

4.4 Visual overview

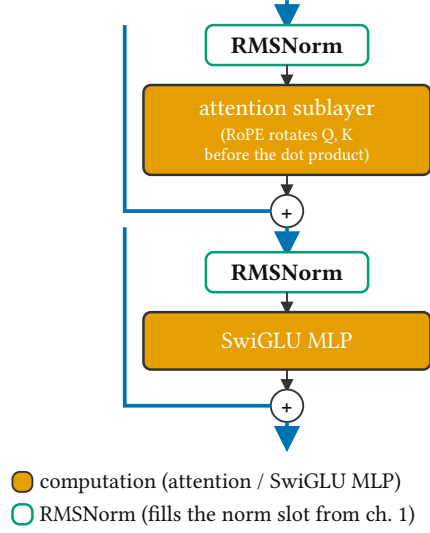


Figure 5: The same residual-stream skeleton from Chapter 1, now filled in with the specific components a modern decoder actually uses: RMSNorm in the norm slots, RoPE applied inside attention (not as a separate block), and a SwiGLU-gated MLP in the feed-forward slot.

4.5 Technical core

RMSNorm. Chapter 1 left **Norm** in Equation 2 unspecified. The original Transformer used LayerNorm, which both re-centers (subtracts the mean) and re-scales (divides by the standard deviation) a vector, then applies a learned scale and a learned bias. RMSNorm drops both the mean-subtraction *and* the learned bias, keeping only a rescale by root- mean-square magnitude:

$$\text{RMSNorm}(x) = \frac{x}{\text{RMS}_\epsilon(x)} \odot \gamma, \quad \text{RMS}_\epsilon(x) = \sqrt{\frac{1}{d_{\text{model}}} \sum_{k=1}^{d_{\text{model}}} x_k^2 + \epsilon} \quad (7)$$

where $x \in \mathbb{R}^{d_{\text{model}}}$ is one position’s vector, $\gamma \in \mathbb{R}^{d_{\text{model}}}$ is a learned per-dimension scale, $\odot$ is elementwise multiplication, and ϵ is a small fixed constant (not learned) added purely to keep the denominator away from zero when a vector’s RMS magnitude is near zero – it has no other role in the formula. *In words:* measure the vector’s overall magnitude (RMS), divide it out so the vector has unit RMS, then let a learned per-dimension scale stretch or shrink each dimension back. There is no learned shift and no mean-subtraction step at all.

Common misconception

“RMSNorm and LayerNorm are identical except for naming.” They are not. RMSNorm drops mean-centering (the $x - \text{mean}(x)$ step) *and* the learned bias term, not just the bias. Dropping mean-centering is the more consequential change: it means RMSNorm never asks “is this vector’s average component large or small,” only “is this vector’s overall magnitude large or small.” Both choices are motivated by the same practical benefit – fewer operations per normalization call, at a cost most current large decoders have judged acceptable – but “fewer operations” and “identical except for naming” are very different claims.

Where normalization sits. Equation 2 wrote **Norm** as happening *before* handing the vector to a sublayer – that placement is called **pre-norm**. The original Transformer instead normalized *after* the sublayer’s output was added back onto the stream (**post-norm**). A third option, **sandwich-norm**, applies normalization in both places around the same sublayer.

Table 3: Pre-norm vs. post-norm vs. sandwich-norm: what changes is placement, not the normalization formula itself

Placement	Where norm sits	Typical motivation
Pre-norm	Before the sublayer reads the stream	Training stability at large depth: gradients flow through an unnormalized residual path all the way back
Post-norm	After the sublayer’s output is added to the stream	The original placement; keeps the stream itself bounded, at some training-stability cost as depth grows
Sandwich-norm	Both before and after the sub-layer	Attempts to recover some of post-norm’s boundedness without losing pre-norm’s stability

Same sublayer, same residual add -- only where "norm" sits changes

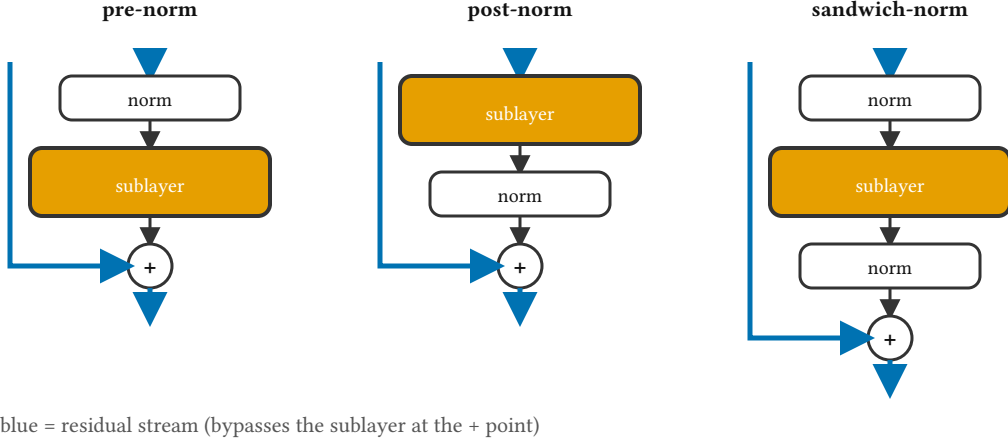


Figure 6: The same sublayer and the same residual addition in all three panels – only where the norm box sits changes.

Working approximation

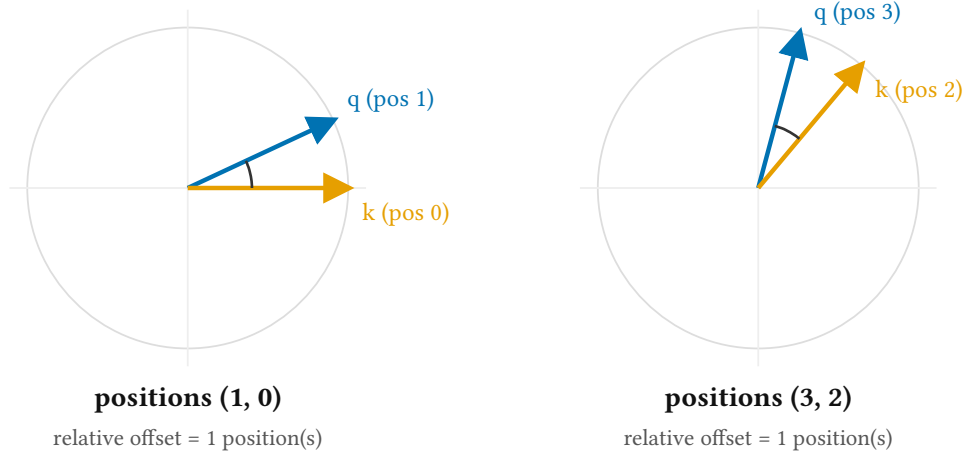
“Pre-norm is more stable to train at depth” is a widely-repeated, well-supported rule of thumb, not a proof that pre-norm dominates post-norm on every axis. Pre-norm’s stability comes with a representational-capacity cost as depth grows, which is part of why sandwich-norm exists at all – if pre-norm were unconditionally better, there would be no reason to pay for a second normalization step. Treat “which placement is best” as a genuine, config-dependent tradeoff, not a settled question.

RoPE. Positional information has to enter the model somewhere, since attention itself (Equation 4) treats its inputs as an unordered set – nothing in $QK^\top$ depends on token order unless something makes it depend on order. Rotary position embeddings (RoPE) inject position by rotating each query and key vector by an angle proportional to its position, *before* the dot product in Equation 4:

$$q'_m = R_m q_m, \quad k'_n = R_n k_n \quad (8)$$

where $q_m, k_n \in \mathbb{R}^{d_{\text{head}}}$ are a query/key pair at positions m and n , and R_m is a rotation matrix built from angles $m\theta_1, m\theta_2, \dots$ that grow with position m (in practice d_{head} is split into 2D pairs, each pair rotated at its own fixed frequency θ_j ; Figure 7 shows one such pair). The reason this matters: rotating *both* vectors by an amount tied to their own absolute position leaves the *angle between them* – and therefore their dot product $q'_m{}^\top k'_n$ – dependent only on the relative offset $m - n$, not on m and n individually.

Same relative offset, different absolute positions -> same angle between q and k



Both panels: the angle traced between q and k is identical, because each vector is rotated by its OWN position, so only the difference in position survives.

Simplified model: real RoPE rotates several 2D subspaces at different frequencies at once; this figure shows one subspace at one frequency to keep the geometry checkable by eye.

Figure 7: Two different absolute-position pairs, (1,0) and (3,2), both with a relative offset of 1 – notice the angle traced between the rotated q and k vectors is the same in both panels.

Common misconception

“RoPE adds a learned position vector to every token embedding.” It does not add anything to the embedding, and it is not learned. RoPE *rotates* the query and key vectors, inside the attention computation, by a fixed (not learned) angle that depends on position. The embedding table from Chapter 1 (Equation 1) is untouched – RoPE’s effect is purely inside Equation 4, applied fresh at every layer that has attention, not once at the input.

Gated feed-forward networks and SwiGLU. Chapter 1 described the feed-forward sublayer as “a small two-matrix network,” deferred until now. A plain feed-forward layer is:

$$\text{FFN}_{\text{plain}}(x) = \sigma(xW_1)W_2 \quad (9)$$

with $W_1 \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}$, $W_2 \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$, and σ an elementwise activation. d_{ff} (the **expansion dimension**) is typically several times wider than d_{model} . A **gated** feed-forward layer, of which SwiGLU is the specific variant most current open decoders use, adds a second, parallel projection that gates the first:

$$\text{SwiGLU}(x) = (\text{Swish}(xW_1) \odot (xW_3)) W_2 \quad (10)$$

with $W_1, W_3 \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}$ and $W_2 \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$ – **three** weight matrices where the plain version had two. *In words:* one branch (W_1 , activated) acts as a gate; a second, purely linear branch (W_3) carries a value; the two are multiplied elementwise before the shared output projection W_2 . Nothing here makes the layer

sparse the way a mixture-of-experts layer is (Chapter 5) – every weight in W_1 , W_2 , and W_3 is used for every token; SwiGLU is a different, denser computation per token, not a routing mechanism.

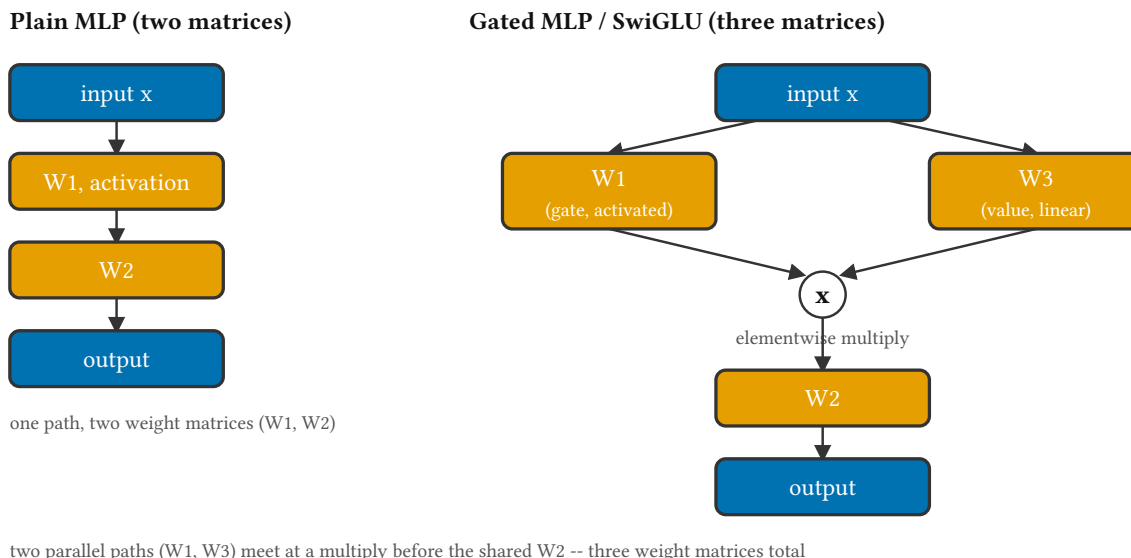


Figure 8: A plain two-matrix MLP (left) against a gated MLP (right): the extra matrix creates a second parallel branch that multiplies against the first, it does not add a third sequential step.

Common misconception

“SwiGLU makes the feed-forward block sparse like an MoE layer.” It does not. Every parameter in a SwiGLU feed-forward layer participates in every token’s forward pass, exactly like the plain version – the only change is *within* one dense computation (an extra multiplicative branch), not *which* parameters get used. Sparsity, in the sense Chapter 5 will use the word, means a router activates only a subset of available parameters per token. SwiGLU has no router and activates 100% of its own weights every time, same as a plain FFN.

Tied vs. untied embeddings. The input embedding table E from Equation 1 and the output head W_{out} from Equation 6 have transposed shapes ($V \times d_{\text{model}}$ and $d_{\text{model}} \times V$) – some models set $W_{\text{out}} = E^T$ (**tied** embeddings, one shared table doing both jobs), others learn two separate matrices (**untied**). Tying halves that specific parameter cost, at the cost of forcing the same table to serve two different roles (looking a token up vs. scoring how token-like a hidden vector is).

Implementation-dependent detail

Whether a given model ties its embeddings is a genuine implementation choice that varies across model families, and this workbook does not have a primary source specifically dedicated to *when* tying is preferred – a tracked sourcing gap (see the Build and Version Note). Treat “tied embeddings” as a real, checkable fact about any specific model’s config – not as a claim this workbook makes a historical or performance argument about.

Convention vs. requirement, applied. Of everything in this chapter, RoPE-vs-some-other-positional-scheme and tied-vs-untied-embeddings are conventions: a model is a well-defined, functioning decoder either way, and real, current models make both choices differently. Pre-norm-vs-post-norm is closer to a requirement-shaped tradeoff at large depth – it is not free to ignore once L is large, even though small models tolerate either. RMSNorm vs. LayerNorm is closest to a pure convention: it changes cost, not what the model is capable of expressing.

Common misconception

“Every modern decoder uses exactly the same block ordering.” It does not. This chapter’s components (RMSNorm, pre-norm, RoPE, SwiGLU) describe a common, current combination, not a single mandatory recipe. Norm placement alone has three real variants in production models (Table 3); tying is a per-model choice; some models still use post-norm. “Modern decoder” describes a family of closely related designs, not one design.

4.6 Worked example

Parameter cost of gating. Verified programmatically (see the Build and Version Note), with $d_{\text{model}} = 8$ and biases excluded from the count (a stated assumption):

Table 4: Gating adds a third matrix – at the same expansion width, that’s a flat 1.5x parameter cost, which is exactly why gated models typically shrink d_{ff} instead of accepting it

Design	d_{ff}	Formula	Parameters
Plain MLP	32	$2 d_{\text{model}} d_{\text{ff}}$	512
Gated MLP, same d_{ff}	32	$3 d_{\text{model}} d_{\text{ff}}$	768 (1.5x plain)
Gated MLP, compensated d_{ff}	21	$3 d_{\text{model}} d_{\text{ff}}$	504 (0.98x plain)

Holding d_{ff} fixed, gating is exactly 1.5x the plain design’s parameters, because the formula gains one more $d_{\text{model}} \times d_{\text{ff}}$ term (3 matrices instead of 2, all the same size). Rather than accept that 1.5x, many real gated-MLP models shrink d_{ff} (here, to about $\frac{2}{3}$ of the plain design’s width) to land back near the plain design’s total parameter count. Both rows are *implementation choices* – nothing about SwiGLU forces either one – but the 1.5x number is a fixed, checkable consequence of adding a third same-sized matrix, not a rough estimate.

4.7 Compare and contrast

Table 5: Two ways to inject positional information, compared on what they cost and how they behave outside the training distribution

Aspect	Learned absolute positional embedding	RoPE
What it encodes	Each position gets its own learned vector	A rotation angle, applied inside attention, derived from position
Extrapolation to longer sequences than trained on	Poor – positions beyond the trained range have no learned vector at all	Better, though not unlimited – the rotation formula is defined for any position, but very long offsets still stress the model in practice
Parameter cost	One full vector per position, up to a fixed max length	None – RoPE has no learned parameters of its own

4.8 Check your understanding

Full solutions are in the answer key (Section 11):

1. **Calculation.** For a query/key pair at positions 1 and 0 (relative offset 1), and a second pair at positions 3 and 2 (also relative offset 1), what does RoPE guarantee about the two pairs' dot products, and what does it *not* guarantee? _____
2. **Compare.** Name the two operations RMSNorm omits relative to LayerNorm. _____
3. **Explain.** Why does a gated feed-forward layer need three weight matrices instead of two – what does the third matrix do that the first one doesn't already? _____
4. **Calculation.** Using the formulas in this chapter, what is the parameter count of a plain MLP with $d_{\text{model}} = 16$, $d_{\text{ff}} = 64$? _____
5. **Misconception check.** Is pre-norm's popularity mainly about training stability or mainly about final model quality? _____
6. **Misconception check.** True or false, and why: "RoPE adds a learned position vector to every token embedding." _____
7. **Misconception check.** True or false, and why: "SwiGLU makes the feed-forward block sparse like an MoE layer." _____

(Space intentionally left for written answers if you are using this as a workbook exercise, not only a reading.)

4.9 Applied exercise

Here is a (fictional, illustrative) model config excerpt:

```
hidden_size: 2048
num_attention_heads: 16
intermediate_size: 5504
hidden_act: "silu"
```

```
rms_norm_eps: 1e-5
tie_word_embeddings: false
rope_theta: 10000.0
```

From this excerpt alone: (a) which normalization scheme does this model use, and how do you know from a field name alone versus what you’d need to check the model’s own code to be sure? (b) does it use a gated feed-forward layer – what field is the giveaway? (c) is `intermediate_size` closer to $4 \times \text{hidden_size}$ (a plain-MLP-style ratio) or closer to $\frac{2}{3} \times 4 \times \text{hidden_size}$ (a gated, compensated ratio, per this chapter’s worked example)? (d) are embeddings tied? Write your reasoning for each, not just the answer – this is the same skill Chapter 6’s case studies will ask of you at real-model scale.

4.10 Interview lens

“Why does this model use RoPE instead of learned positional embeddings?” A strong answer states RoPE’s relative-position property (Equation 8) as the mechanism, names at least one concrete consequence (no learned positional parameters; different, generally better long-context extrapolation behavior than a learned table with a fixed max length), and resists the temptation to just say “it’s more modern” without saying what specifically changed. *“Walk me through what SwiGLU changes about the MLP”* is answered well by describing the extra parallel branch and the elementwise multiply (Equation 10), explicitly including the 1.5x-or-compensated parameter tradeoff from this chapter’s worked example – an answer that skips the parameter cost entirely is incomplete for a systems-flavored interview.

4.11 Chapter recap

- RMSNorm drops both mean-centering and the learned bias LayerNorm has, keeping only a magnitude rescale (plus a small fixed ϵ to avoid dividing by zero).
- Pre-norm, post-norm, and sandwich-norm describe *where* normalization sits relative to a sublayer and the residual add – not three different normalization formulas.
- RoPE rotates queries and keys by a position-dependent angle inside attention, so their dot product depends only on relative offset – it adds nothing to the embedding table and learns nothing.
- A gated (SwiGLU-style) feed-forward layer adds one parallel matrix and one elementwise multiply; at matched expansion width that’s a flat 1.5x parameter cost, which is why real models often shrink the expansion width to compensate.
- Tied vs. untied embeddings and RoPE-vs-alternative are conventions; norm placement is closer to a real training-stability tradeoff at depth. None of this chapter’s components is used identically by every current model.

4.12 Sources and further reading

Primary sources for this chapter: [3] for RoPE, [4] for RMSNorm, [5] for SwiGLU/gated feed-forward layers. Secondary/ supporting: [2], used for the norm-placement comparison framing and to cross-check which real model families use which combination of these components; this source is actively updated, so treat any model-specific detail drawn from it as current as of its last verification (see the Build and Version Note), not timeless. This chapter works around two tracked sourcing gaps: tied embeddings has no dedicated primary source, and the sandwich-norm claim leans on a secondary source rather than a model’s own technical report (both noted in the Build and Version Note).

5 Chapter 3: Attention Head Structure and Cache-Efficient Variants

Why Fewer Key/Value Heads Means a Smaller Cache – Not a Smaller Model

5.1 Learning objectives

By the end of this chapter you should be able to:

- Distinguish query-head count (H_q) from key/value-head count (H_{kv}), and write $Q/K/V$'s shapes before and after splitting into heads.
- Explain how multiple query heads can share a key/value head, and distinguish MHA, MQA, and GQA structurally on that one axis.
- Calculate a decoder's logical KV-cache size under stated assumptions, and state which projection parameters change when H_{kv} changes (and which don't).
- Explain why reducing H_{kv} lowers cache memory and memory traffic without reducing every part of attention computation by the same factor.
- Identify tradeoffs among MHA, MQA, and GQA without claiming any one is universally best, and distinguish an architecture-level cache calculation from measured serving-memory usage.

5.2 Why this matters

Chapters 1-2 gave every block one query head per key/value head, without saying that was a choice. It is. The number of key/value heads a model uses is an independent design knob from the number of query heads, and it is the single knob most responsible for how much memory autoregressive decoding needs per active sequence. Get this chapter's distinction wrong and two very different mistakes follow: overstating what changes (claiming a smaller cache means a proportionally smaller or faster model) or understating it (missing that cache memory, not FLOPs, is usually the actual serving bottleneck this chapter is about).

5.3 Mental model or analogy

Analogy

Queries are readers asking different questions; keys are index entries used to decide where to look; values are the information retrieved at that index entry. Standard attention (MHA) gives every reader their own private index and information store. MQA makes all readers share one shared index and store. GQA gives each *group* of readers a shared index and store – a middle ground between the two.

Where this breaks down: keys and values are learned vector projections, not literal documents. Sharing a key/value head changes how much distinct *representational capacity* the model has to work with, not merely how many copies of a document are stored – two readers sharing an index can no longer each shape that index around only their own question.

5.4 Visual overview

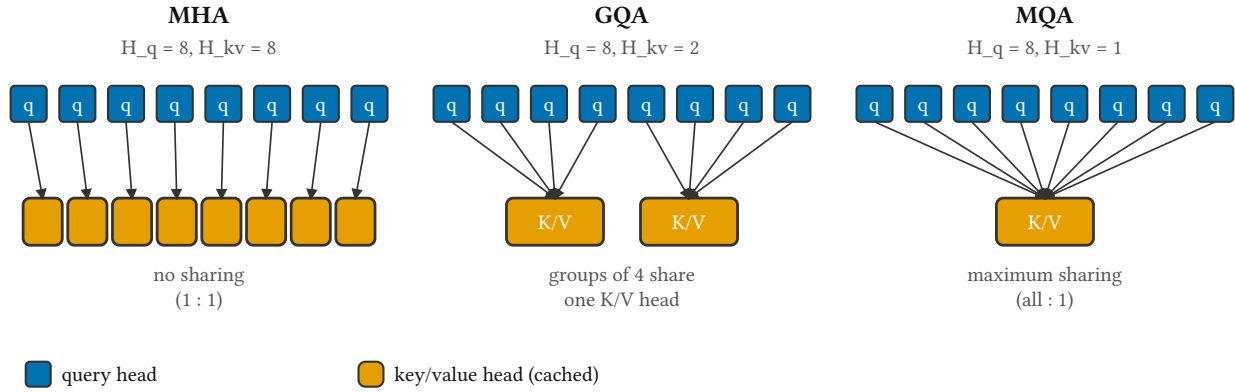


Figure 9: MHA, GQA, and MQA shown with the same $H_q = 8$ query heads in every panel – nothing removes a query head. What changes is how many key/value heads those query heads fan into: 8 (no sharing), 2 (groups of 4 share one), or 1 (all share one). Source: mechanism per [6], [7].

5.5 Technical core

This chapter uses H_q (query heads), H_{kv} (key/value heads), d_{head} (head dimension, with $d_{\text{model}} = H_q d_{\text{head}}$), S (tokens retained in the cache), T_q (query positions processed in the *current* step – $T_q = S$ during prefill, $T_q = 1$ for an ordinary decode step), L (layers), and `bytes_per_elem` (storage width per cached number, e.g. 2 for bf16). Logical shapes, head axis explicit (physical memory layouts vary by implementation):

$$Q : (B, T_q, H_q, d_{\text{head}}), \quad K : (B, S, H_{kv}, d_{\text{head}}), \quad V : (B, S, H_{kv}, d_{\text{head}}) \quad (11)$$

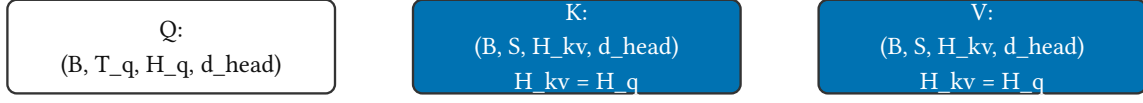
MHA, MQA, GQA: one axis, three points. Standard multi-head attention (MHA, [1]) sets $H_{kv} = H_q$: every query head has its own key/value head. Multi-query attention (MQA, [6]) sets $H_{kv} = 1$: every query head shares the same single key/value head. Grouped-query attention (GQA, [7]) sets $1 < H_{kv} < H_q$: query heads are split into H_q/H_{kv} equal groups (this simple equal-group presentation requires H_{kv} to divide H_q), each group sharing one key/value head. Figure 9 shows all three at $H_q = 8$.

Common misconception

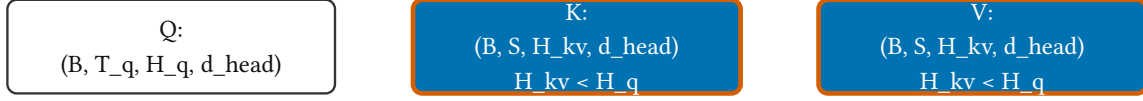
“GQA reduces the number of query heads.” It does not. H_q is untouched by any of these three variants – Figure 9 draws the same 8 query heads in every panel. Only H_{kv} changes.

What stays fixed, what changes. Across MHA/MQA/GQA at the same H_q and d_{head} : the query projection’s shape, the per-head attention *output* shape, the concatenated output width, and the output projection width are all unchanged – every one of them depends on H_q , not H_{kv} . What changes is the K/V projections’ output width and, per position, how many K/V vectors get stored. Figure 10 marks the one shape that differs.

MHA



GQA



Q's shape never changes between MHA and GQA -- only K/V's head axis (outlined in red) shrinks.

Group size = H_q / H_{kv} (e.g. H_q=8, H_{kv}=2 -> each K/V head serves a group of 4 query heads).

Figure 10: Only K/V 's head axis (H_{kv}) shrinks between MHA and GQA; Q 's shape is untouched.

Projection parameters. Ignoring biases, per layer:

$$P_Q = d_{\text{model}} H_q d_{\text{head}}, \quad P_K = d_{\text{model}} H_{kv} d_{\text{head}}, \quad P_V = d_{\text{model}} H_{kv} d_{\text{head}}, \quad P_O = H_q d_{\text{head}} d_{\text{model}} \quad (12)$$

Only P_K and P_V depend on H_{kv} ; P_Q and P_O depend only on H_q . Shrinking H_{kv} shrinks two of these four terms, not all four – the worked example below turns this into a concrete ratio.

KV-cache size. The quantity these mechanisms actually optimize is not parameter count but cache memory:

$$M_{KV} = 2 B S L H_{kv} d_{\text{head}} \text{ bytes_per_elem} \quad (13)$$

where the factor 2 counts keys *and* values. This is a **logical, theoretical minimum** – per AGENTS.md's distinction between logical and measured memory, it explicitly excludes allocator overhead, block/page metadata, fragmentation, temporary attention workspaces, framework buffers, duplicated or padded storage, sharding-specific per-device layouts, prefix-sharing effects, and any cache quantization not already reflected in `bytes_per_elem`.

Common misconception

“The KV-cache formula gives exact observed GPU memory usage.” It does not. Equation 13 is a logical minimum computed from a config; what `nvidia-smi` or a profiler reports also includes every exclusion listed above, and will always be larger.

Decode-time computation. At each decode step, T_q new query positions are computed (ordinarily $T_q = 1$; chunked prefill and speculative-decoding verification – both later topics, named here only as forward references – can make $T_q > 1$). Every one of the H_q query heads still produces its own attention scores; a shared key/value head is simply reused (logically broadcast) across every query head in its group. Reducing H_{kv} principally reduces what gets *stored* and *moved* – it does not reduce the number of query-side dot products, so it is not accurate to say GQA makes attention some fixed multiple faster; the win this chapter is about is memory and memory-bandwidth, not FLOPs.

5.6 Worked example

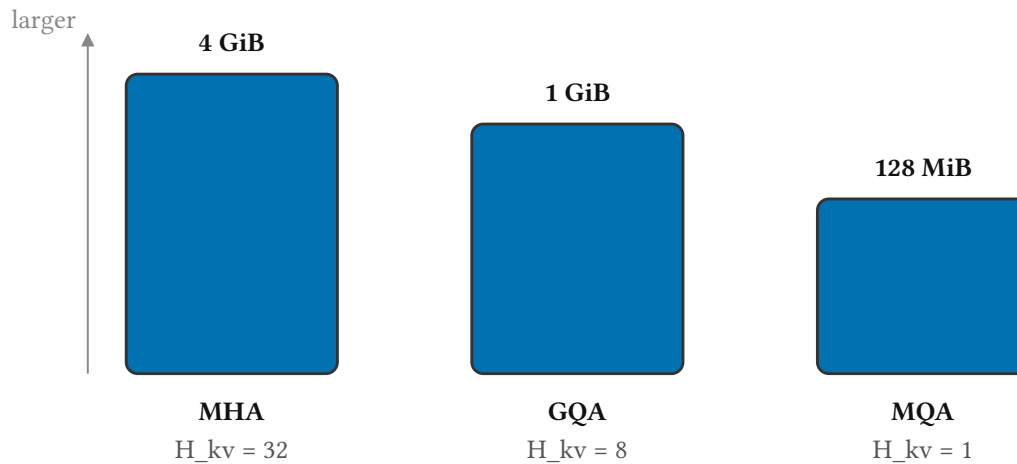
KV-cache memory. Base config: $L = 32$, $H_q = 32$, $d_{\text{head}} = 128$ (so $d_{\text{model}} = 4096$), $S = 8192$, $B = 1$, bf16 (bytes_per_elem = 2) – verified programmatically (see the Build and Version Note):

Table 6: KV-cache size for one sequence at $S = 8192$; the reduction factor equals $H_{kv, \text{variant}} / H_{kv, \text{MHA}}$ exactly

Variant	H_{kv}	Equation 13 result	vs. MHA
MHA	32	4,294,967,296 B = 4 GiB	1x
GQA	8	1,073,741,824 B = 1 GiB	0.25x (4x smaller)
MQA	1	134,217,728 B = 128 MiB	0.03125x (32x smaller)

Logical KV-cache size per sequence (L=32, H_q=32, d_head=128, S=8192, B=1, bf16)

bar height is log2-scaled for visibility -- printed values are the actual linear byte counts



Logical minimum only -- excludes allocator overhead, page metadata, fragmentation, attention workspaces, and framework buffers. Not a measured GPU-memory number.

Figure 11: The same three numbers as Table 6 – bar height is log2-scaled so the smallest bar stays visible; the printed values are the actual linear byte counts.

Cache size scales *linearly* in each of B , S , L , H_{kv} , and bytes_per_elem – e.g. at $B = 16$ active sequences instead of 1, every number in Table 6 simply multiplies by 16 (4 GiB → 64 GiB for MHA, 1 GiB → 16 GiB for GQA, 128 MiB → 2 GiB for MQA); nothing about the formula’s *shape* changes.

Projection parameters, same config. Per layer, from Equation 12:

Table 7: Per-layer attention parameters for the same base config – $K+V$ shrinks by exactly the cache-reduction ratio, but the complete attention block (which also includes Q and O , both fixed) shrinks by much less

Variant	P_Q	P_K+P_V	P_O	Total ($Q+K+V+O$)	$K+V$ vs. MHA	Total vs. MHA
MHA	16,777,216	33,554,432	16,777,216	67,108,864	1x	1x
GQA	16,777,216	8,388,608	16,777,216	41,943,040	0.25x	0.625x
MQA	16,777,216	1,048,576	16,777,216	34,603,008	0.03125x	0.516x

GQA’s $K+V$ parameters drop by exactly 4x, matching its cache reduction – but the *complete* attention block (adding the unchanged P_Q and P_O) drops by only 1.6x (to 0.625 of MHA’s total), and MQA’s 32x $K+V$ reduction becomes less than a 2x reduction (0.516x) once Q and O are included. This is Equation 12’s point made concrete: K/V savings do not translate one-for-one into whole-model or whole-attention-block savings.

Common misconception

“If GQA cuts KV-cache memory by 4x, the whole model becomes 4x smaller.” Table 7 shows why not: the complete attention block’s parameters (which include the unchanged Q and O projections) shrink by only 1.6x here, and the feed-forward sublayers (chapters 1-2) are untouched by H_{kv} entirely.

5.7 Compare and contrast

Table 8: MHA vs. GQA vs. MQA, compared without ranking one as strictly best – each row is a tradeoff, not a verdict

	MHA	GQA	MQA
H_{kv}	$H_{kv} = H_q$	$1 < H_{kv} < H_q$	$H_{kv} = 1$
K/V sharing	none	groups of H_q/H_{kv} query heads	all query heads
Logical KV-cache size	largest (1x)	intermediate (config-dependent)	smallest ($1x/H_q$)
K/V projection params	largest	intermediate	smallest
Representation flexibility	most (every head has its own K/V)	reduced, tunable via group size	least
Implementation considerations	simplest; no grouping logic	requires choosing a group size that divides H_q	simplest sharing case; largest capacity reduction
Common motivation	small models, or when cache memory isn’t the bottleneck	a tunable memory/flexibility tradeoff point	maximum cache/bandwidth savings when acceptable

Common misconception

“MQA is universally superior because it uses the least cache.” Least cache is not the only deployment factor. MQA gives every query head the least representational flexibility of the three (all of them share one key/value head, with no group boundary at all), which is exactly the tradeoff [7] introduces GQA to soften – reducing H_{kv} buys memory at a flexibility cost, and where that cost is acceptable is model- and workload-dependent, not a fixed rule.

5.8 Check your understanding

Full solutions are in the answer key (Section 12):

1. **Explain.** What is the one structural difference between MHA, MQA, and GQA – and what stays identical across all three? _____
2. **Calculation.** For $H_q = 8$ query heads and $H_{kv} = 2$ key/value heads, what is the group size, and how many query heads share each key/value head?

3. **Calculation.** Write the shapes of Q and K (per Equation 11) for $B = 1$, $T_q = 1$, $S = 100$, $H_q = 16$, $H_{kv} = 4$, $d_{\text{head}} = 64$. _____
4. **Quick check.** Which two of P_Q , P_K , P_V , P_O change when H_{kv} changes, holding H_q and d_{head} fixed?

5. **Misconception check.** True or false, and why: “GQA reduces the number of query heads.”

6. **Misconception check.** True or false, and why: “if GQA cuts a model’s KV-cache memory by 4x, the whole model becomes 4x smaller.” _____
7. **Calculation.** For $L = 24$, $H_q = 16$ (all used as H_{kv} , i.e. MHA), $d_{\text{head}} = 128$, $S = 4096$, $B = 1$, bf16: compute the logical KV-cache size in bytes and MiB using Equation 13.

(Space intentionally left for written answers if you are using this as a workbook exercise, not only a reading.)

5.9 Applied exercise

A serving workload needs long contexts and high concurrency. Candidate model A uses MHA; candidate B uses GQA, both otherwise the same size and S . Using Equation 13, compare their per-sequence cache cost at your own chosen (long) S and (large) B , then explicitly name at least one deployment factor besides cache memory (e.g. representation flexibility, implementation maturity, quality at the model’s specific task) that could still favor the MHA candidate despite its larger cache. A serving decision is not a cache-size comparison alone.

5.10 Interview lens

“Walk me through why GQA reduces KV-cache cost, and tell me what it does not automatically make cheaper.”
A strong answer names H_{kv} (not H_q) as the changing quantity, states Equation 13’s linear dependence on H_{kv} , and explicitly names what does *not* shrink proportionally: the query and output projections (P_Q , P_O), the number of query-side attention computations, and (per Table 7) the complete attention block’s total parameter count. An answer that only says “it uses less memory” without naming what stays fixed is incomplete for a systems-flavored interview.

5.11 Chapter recap

- H_q and H_{kv} are independent: MHA sets $H_{kv} = H_q$, MQA sets $H_{kv} = 1$, GQA sets $1 < H_{kv} < H_q$ – the same head-sharing axis at three points, never fewer query heads.
- Only the K/V projections’ shape and the KV-cache size depend on H_{kv} ; Q ’s shape, the output projection, and the number of query-side attention computations depend on H_q alone.
- The KV-cache formula ($2 \times B \times S \times L \times H_{kv} \times d_{\text{head}} \times \text{bytes_per_elem}$) is a logical minimum, not a measured GPU-memory number.
- K/V parameter and cache savings do not translate one-for-one into whole-model or whole-attention-block savings, and reducing H_{kv} does not proportionally cut attention FLOPs.
- None of MHA, GQA, or MQA is universally best – each trades cache memory against representational flexibility differently.

5.12 Sources and further reading

Primary sources for this chapter: [1] for the MHA baseline, [6] for MQA, [7] for GQA (including its up-training recipe for converting an existing MHA checkpoint, reported at roughly 5% of original pretraining compute – their own reported figure, not a universal constant). Secondary/supporting: [2] for cross-model framing, [8] for the KV-cache memory formula’s systems framing. A further latent-compression technique that shrinks the cache by a different mechanism than head-sharing exists and is left for a later chapter’s case studies, not covered here.

6 Chapter 4: Reducing Attention Cost: Windows, Sparsity, and Latent KV Representations

Three Different Problems, Not One Knob

6.1 Learning objectives

By the end of this chapter you should be able to:

- Explain why full causal attention grows expensive as retained context grows, without conflating score-matrix cost, KV-cache storage, total model FLOPs, wall-clock latency, memory bandwidth, and temporary workspace.
- Distinguish *attending to fewer positions* from *storing a smaller representation per position* as independent axes, and explain why sparse attention does not automatically shrink the KV-cache.
- Draw sliding-window attention’s receptive field across stacked layers, and state the conceptual purpose of occasional global layers or tokens.
- Compare GQA and MLA structurally (GQA shares K/V heads; MLA compresses what is stored per token) and state what a DeepSeek-style MLA cache retains, including why a separately-retained rotary component is needed.
- Explain why an optimized MLA implementation need not reconstruct full K/V tensors every decode step, and recognize recurrent/ linear-state alternatives as a third design family without claiming this chapter teaches their mechanics.

6.2 Why this matters

Chapter 3 fixed *how many* query heads share a key/value head. This chapter is about a different question entirely: given a fixed head structure, what else can shrink attention’s cost as retained context grows? Three genuinely different strategies exist – attend to fewer positions, store less per retained position, or replace the growing history with something that doesn’t grow – and they solve different problems. Treating them as interchangeable, or as three strengths of one dial, is the single biggest misreading this chapter is built to prevent.

6.3 Mental model or analogy

Think of the retained context as a shelf of books a reader might need again. Strategy 1 (fewer positions) means only checking nearby shelves. Strategy 2 (smaller representation) means keeping one compressed summary card per book instead of the whole book. Strategy 3 (replace growing state) means keeping a running summary that never grows, discarding the shelf entirely. A reader who confuses “check fewer shelves” with “keep shorter notes” would misjudge what they can still find later – exactly the confusion this chapter targets.

6.4 Visual overview

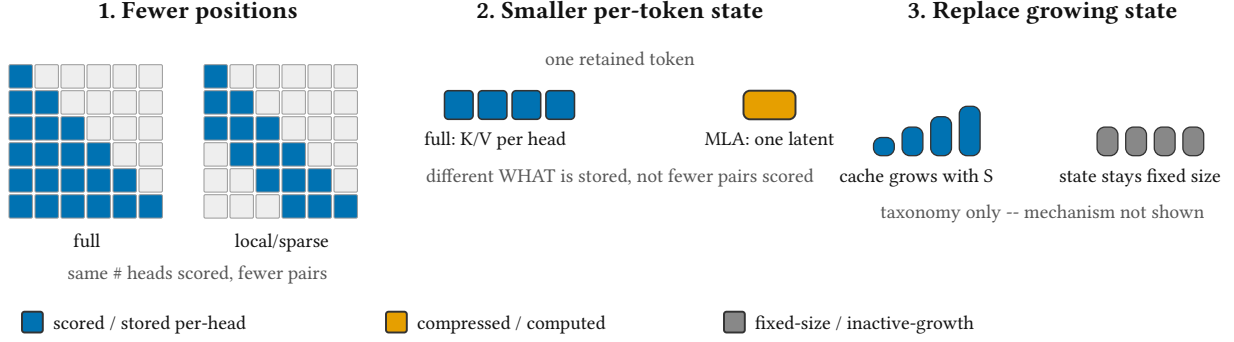


Figure 12: Three different axes for reducing attention's cost, not one continuum: which positions are scored, what is stored per retained position, and whether the history grows at all. Panel 3 names the third family without teaching its mechanics.

6.5 Technical core

Full attention's baseline. For a sequence of retained length S , the attention score matrix has S^2 entries per head – full attention costs $O(S^2)$ score computations for a whole sequence. During incremental decoding, each single new query attends over all S retained positions, so per-step attention work grows with S ; summed over S decode steps this totals $O(S^2)$ as well. The **KV-cache**, by contrast, grows *linearly* in S (per Equation 13 from Chapter 3) – storage and score-computation are different growth curves, and neither is the same as total model FLOPs (which includes the feed-forward sublayers), wall-clock latency (hardware- and batch-dependent), memory bandwidth (the rate data moves, not the amount), or temporary softmax/attention workspace (excluded from the cache formula entirely).

Sliding-window attention. Each position attends only to a bounded window of W recent positions – the *direct* receptive field is W , regardless of S . Stacking layers extends this indirectly:

$$\text{effective receptive field after } L \text{ stacked windowed layers} = L \times W \quad (14)$$

For $W = 4$ over 3 stacked layers, the direct window per layer stays 4, but the effective reach grows to $3 \times 4 = 12$ (verified programmatically) – wider than one layer's window, but still not unrestricted global attention.

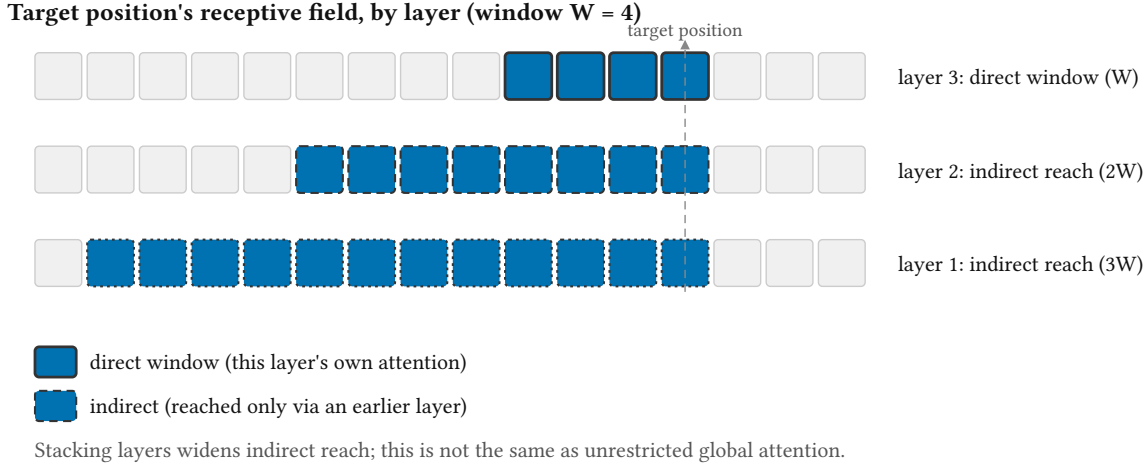


Figure 13: A target position’s direct window is narrow at any one layer; tracing back through earlier layers, the region of positions that can indirectly influence it widens by W per layer – distinct from unrestricted global attention.

Figure 13 marks direct vs. indirect reach explicitly. If a layer strictly evicts state outside its window, that layer’s cache growth can be bounded; real serving systems may still retain more state than the strictest bound, depending on implementation. Real models often mix strategies: most layers windowed, a minority full (“global”) layers, or selected global tokens visible to every position – recovering long-range access without paying full attention’s cost everywhere. Mistral 7B’s window-4096 design with a rolling-buffer cache is one deployed example, reporting roughly an 8x cache reduction at 32K context versus full attention ([9]).

Sparse attention. A sparse pattern (e.g. strided/block factorizations, [10]) restricts *which* query-key pairs are scored, reducing $O(S^2)$ score computation. This is a different statement from what must stay *cached*: some patterns permit dropping positions no future query will ever revisit, but others (e.g. a strided pattern where a future layer or a future token’s stride offset can reach any past position) must retain every token’s K/V regardless, because some future query may still access it.

Common misconception

“Sparse attention automatically means a proportionally smaller KV cache.” It does not. Sparsity is a statement about which pairs get *scored*; the cache must hold whatever a *future* query might still need, which depends on the specific pattern, not sparsity alone.

Multi-head latent attention (MLA). GQA (Chapter 3) shares whole K/V *heads* across query groups – the per-head representation is unchanged, just fewer distinct copies. MLA changes *what* is stored per token: DeepSeek-V2 ([11]) jointly compresses K and V into one shared latent $c_t^{KV} \in \mathbb{R}^{d_c}$ per token per layer, with $d_c \ll H_q d_{\text{head}}$, via a down-projection; per-head K/V are then *derived* from that latent when needed, not stored directly. RoPE complicates this: rotating the compressed latent directly would tie the up-projection to a position-specific rotation matrix that cannot later be folded elsewhere, forcing every prior token’s key to be recomputed at each step. MLA’s fix retains a small, separately-cached decoupled rotary component (d_{rope} , shared across heads) alongside the latent, so:

$$M_{\text{MLA}} = B S L (d_c + d_{\text{rope}}) \text{ bytes_per_elem} \quad (15)$$

Equation 15 and Chapter 3’s M_{KV} describe **different cached representations** – a numeric comparison is meaningful only for an explicitly defined architecture, and both remain logical minimums (runtime overhead excluded, same exclusions as Chapter 3).

Common misconception

“MLA is just GQA with fewer KV heads.” It is not. GQA shares whole per-head K/V representations across query groups; MLA compresses K and V *jointly* into one smaller latent representation per token, a different mechanism, not a further point on GQA’s head-count axis.

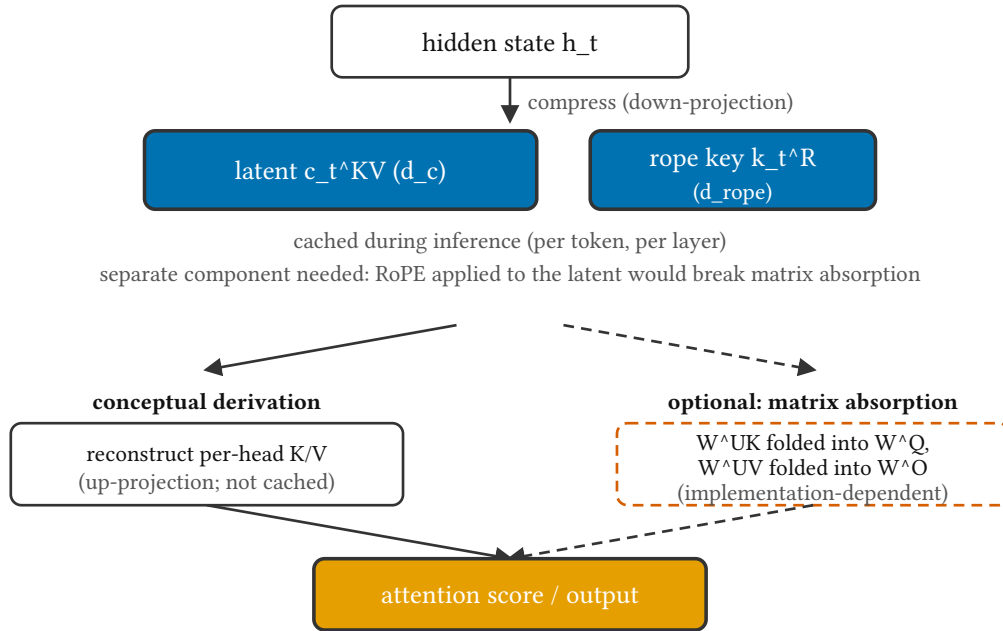


Figure 14: MLA’s cache pathway: only the compressed latent and the decoupled rotary component are cached. Reconstructing full K/V (left) is always a conceptually valid derivation; an optimized implementation may instead take the dashed, optional matrix-absorption path, which never materializes full K/V at all.

Figure 14 shows what is actually cached and where reconstruction happens. Because a score is $q_t^\top k_j = q_t^\top (W^{UK} c_j^{KV})$, and matrix multiplication reassociates as $(q_t^\top W^{UK}) c_j^{KV}$, an optimized implementation can fold W^{UK} into the query-side projection ahead of time (and, symmetrically, W^{UV} into the output projection) – so it never has to materialize full K/V at all ([11]). This is an available optimization, not a mandatory step every implementation takes; the conceptual derivation (reconstruct, then attend) remains valid either way.

6.6 Worked example

Sliding-window reach. $W = 4$, layers 1/2/3 give effective reach 4/8/12 (Equation 14, verified programmatically).

GQA vs. MLA cache. Reusing Chapter 3’s base config ($L = 32$, $H_q = 32$, $d_{\text{head}} = 128$, $S = 8192$, $B = 1$, bf16) and DeepSeek-V2’s own reported ratio ($d_c = 4 d_{\text{head}} = 512$, $d_{\text{rope}} = 0.5 d_{\text{head}} = 64$; [11] Table 1):

Table 9: GQA vs. MLA on one shared base config – verified programmatically; see the Build and Version Note

Representation	Per-sequence cache	vs. MHA (4 GiB)
GQA, $H_{kv} = 8$ (Ch. 3)	1 GiB	0.25x
MLA, $d_c + d_{\text{rope}} = 576$	288 MiB	0.070x

This MLA configuration’s cache equals a GQA config with exactly 2.25 groups ([11]’s own stated equivalence, reproduced here as a verified invariant) – MLA and GQA can land at comparable *sizes* for some configuration, without being the same mechanism. This toy ratio is **not** a claimed speedup: cache compression is not the same as end-to-end latency, model quality and training behavior are outside this arithmetic, and runtime overhead is excluded throughout.

6.7 Compare and contrast

Table 10: Six strategies compared without forcing false binaries – “pattern-dependent” and “implementation-dependent” are answers, not omissions

	Full attn.	Sliding-window	Sparse (generic)	GQA	MLA	Recurrent/linear
Changes connectivity?	–	yes	yes	no	no	yes (different primitive)
Changes stored representation?	–	pattern-dependent	pattern-dependent	yes (fewer copies)	yes (compressed)	yes (fixed-size state)
Can bound retained history?	no	pattern-dependent	pattern-dependent	no	no	yes
Direct access to every prior token?	yes	no (indirect only)	pattern-dependent	yes	yes (via latent)	no
Primary benefit	–	compute + memory	compute	memory/bandwidth	memory/bandwidth	memory + compute
Principal tradeoff	cost grows with S	indirect long-range access	cache not guaranteed smaller	some capacity loss	implementation complexity	mechanics not covered here

6.8 Check your understanding

Full solutions are in the answer key (Section 13):

1. **Quick check.** As retained context S grows, what grows quadratically and what grows linearly?

2. **Explain.** Why must MLA retain a separately-cached rotary component alongside its compressed latent?

3. **Explain.** Does an optimized MLA implementation have to reconstruct full K/V tensors every decode step?

4. **Misconception check.** True or false: a sliding window means information older than W tokens can never affect the current representation. _____
5. **Misconception check.** True or false: compressing the cache by 4x makes decoding 4x faster.

6. **Calculation.** For $L = 24$, $H_q = 16$, $d_{\text{head}} = 128$: using DeepSeek-V2's ratio ($d_c = 4d_{\text{head}}$, $d_{\text{rope}} = 0.5d_{\text{head}}$), compute the MLA cache elements per token and compare to GQA at $H_{kv} = 4$.

(Space intentionally left for written answers if you are using this as a workbook exercise, not only a reading.)

6.9 Applied exercise

A workload needs long context, high concurrency, and occasional retrieval of very old tokens. Explain why a pure sliding window may be insufficient, and what combination of mechanisms (global layers/tokens, GQA, MLA) could help.

6.10 Interview lens

“Compare GQA, sliding-window attention, and MLA. What resource does each primarily reduce, and what does each give up or complicate?” A strong answer names each one's primary win and cost: GQA (memory/bandwidth, at some representational cost), sliding-window (compute and memory, at the cost of indirect long-range access), and MLA (memory/bandwidth, at the cost of implementation complexity) – and does not collapse any two into the same mechanism.

6.11 Chapter recap

- Full attention's score matrix grows $O(S^2)$; its KV-cache grows linearly in S – different curves, never the same claim.
- Attending to fewer positions (windows, sparsity) and storing less per position (MLA) are independent axes; sparse attention does not automatically shrink the cache.
- Sliding-window's direct reach is W ; stacked layers reach $L \times W$ indirectly – not unrestricted global attention.
- MLA compresses K/V jointly into one latent per token (plus a rotary component); GQA shares whole per-head copies instead – different mechanisms, occasionally comparable sizes.
- An optimized MLA implementation may skip explicit K/V reconstruction via matrix absorption, but this is an available optimization, not a mandatory step.
- Recurrent/SSM/linear-attention replace the growing cache with a fixed-size state – named here as a third family, mechanics deferred.

6.12 Sources and further reading

Primary sources: [9] (sliding-window), [10] (sparse attention), [11] (MLA – compression formula, decoupled-RoPE reasoning, and cache-per-token table verified against the paper's full text, Sections 2.1.1-2.1.4

and Table 1, not a snippet). Secondary/supporting: [2]. This chapter keeps recurrent/SSM/linear-attention at the taxonomy level, without a primary source for its mechanics.

7 Chapter 5: Mixture of Experts: More Parameters, Selective Compute

Total, Active, Compute, and Communication Are Four Different Questions

7.1 Learning objectives

By the end of this chapter you should be able to:

- Distinguish a dense feed-forward block from a sparse MoE block, and explain the roles of the router, experts, top- k selection, routing weights, and an optional shared expert.
- Calculate total and active parameters for the routed-expert bank *and* for the complete model, and explain why these are not the same ratio.
- Explain why sparse activation does not make unused expert weights disappear from memory, and describe load imbalance, expert capacity, and what a load-balancing loss penalizes (without guaranteeing balance).
- Distinguish MoE sparsity (which *parameters* activate) from sparse attention (Chapter 4's which *positions* are scored), and explain why expert parallelism introduces device-to-device communication.
- Compare model capacity, computation, memory footprint, and communication as four separate quantities.

7.2 Why this matters

Every mechanism so far has changed *how* a fixed set of parameters is used per token. MoE changes *which* parameters get used at all. Getting this chapter's central distinction wrong – treating “fewer active parameters” as “fewer stored parameters,” or as “proportionally less compute,” or as “no communication cost” – is the single most common MoE misreading, and this chapter is built to prevent exactly that.

7.3 Visual overview

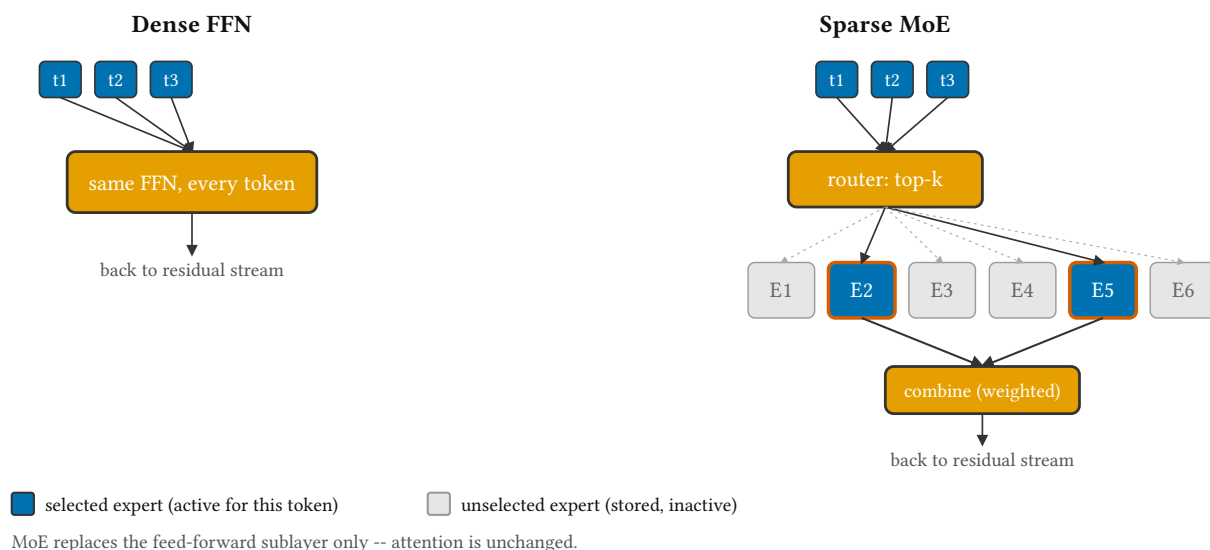


Figure 15: MoE replaces the feed-forward sublayer, not attention: a router selects a subset of experts per token; unselected experts are stored but not used for that token.

7.4 Technical core

From dense to sparse. Chapter 2’s gated feed-forward block sends every token through the *same* parameters. A sparse MoE layer replaces that single block with a **router** that scores E **routed experts**, selects the top- k for each token, runs only those experts, and **combines** their outputs with routing weights before returning the result to the residual stream. Exact routing/combination rules vary by architecture; this chapter’s formulation is representative, not universal.

Routing. For a token representation x :

$$r = W_r x, \quad p = \text{softmax}(r), \quad T(x) = \text{TopK}(p, k) \quad (16)$$

r are the router logits, p the per-expert probabilities, and $T(x)$ the selected top- k index set. The combined output is:

$$y(x) = \sum_{e \in T(x)} \alpha_e(x) \cdot \text{Expert}_e(x) \quad (17)$$

with $\alpha_e(x)$ a routing weight (often, but not always, the normalized p_e). Not every production MoE normalizes weights, handles capacity, or adds auxiliary losses identically.

Total vs. active parameters. With p_e parameters per routed expert, an optional always-active shared-expert path of p_s parameters, and p_{base} for everything else (attention, embeddings, norms, the router):

$$\underbrace{E p_e}_{\text{routed total}}, \quad \underbrace{k p_e}_{\text{routed active}}, \quad \underbrace{p_{\text{base}} + E p_e + p_s}_{\text{model total}}, \quad \underbrace{p_{\text{base}} + k p_e + p_s}_{\text{model active}} \quad (18)$$

E/k is the routed-expert-bank’s *own* ratio – it is **not** the complete model’s total/active ratio, since p_{base} and p_s are active regardless of routing. Published “A-B” model notation (e.g. “236B-21B”) is convention-dependent; check official documentation before assuming what it counts. Active parameters are also not identical to exact FLOP counts (the router itself costs compute too).

Common misconception

“A 64-expert top-2 MoE activates only 2/64 of the whole model.” It does not, unless the statement is explicitly restricted to the routed expert bank. Attention, embeddings, normalization, the router, and any shared expert stay active regardless – Equation 18’s model-level ratio is always closer to 1 than E/k is.

For one token, top-2 of 8 routed experts selected

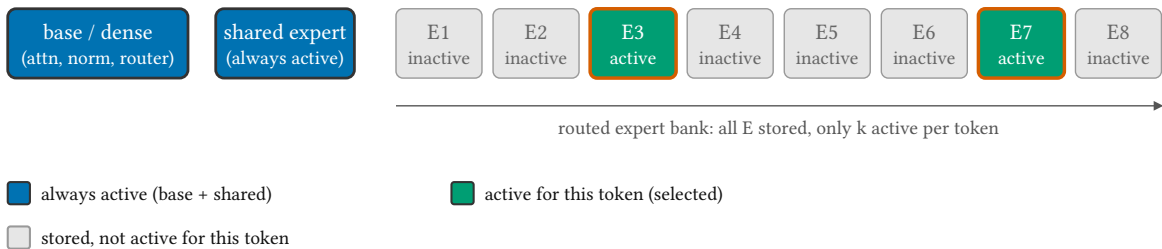


Figure 16: For one token, only k of E routed experts are active; base parameters and any shared expert are active regardless; all E experts remain stored either way.

Common misconception

“Inactive experts consume no memory.” They do. An expert not selected for a given token is still a stored weight matrix that must fit in memory – sparsity is about *compute*, not *storage*.

Load balancing and capacity. Without constraints, many tokens may route to a few popular experts, overloading them while others sit idle – wasting capacity and risking *routing collapse* (reduced specialization). Systems bound this with a **capacity factor** (how many tokens an expert can take before **overflow**, handled by dropping or rerouting, implementation-dependent) and an **auxiliary load-balancing loss** that *encourages*, but does not *guarantee*, balanced routing in any given batch.

Shared experts. Some transformations are useful for most tokens; an always-active shared expert can capture that shared knowledge, reducing pressure on routed experts to duplicate it – DeepSeekMoE adds shared experts while *reducing* routed top- k activation to hold total active compute roughly constant ([12]). Shared experts are not universally necessary; their benefit depends on architecture and training.

Expert parallelism. Routed experts may be split across devices, so tokens must be dispatched to whichever device holds their selected expert and results returned – an **all-to-all** exchange, in plain terms: every device may need to send tokens to, and receive results from, every other device. Uneven routing produces uneven device workloads, and this communication can offset the sparse-compute benefit MoE is chasing. Fewer active expert parameters does **not** mean all expert weights fit on one device – total storage and active computation answer different questions. (Collective-communication algorithms and cluster topology are out of scope here.)

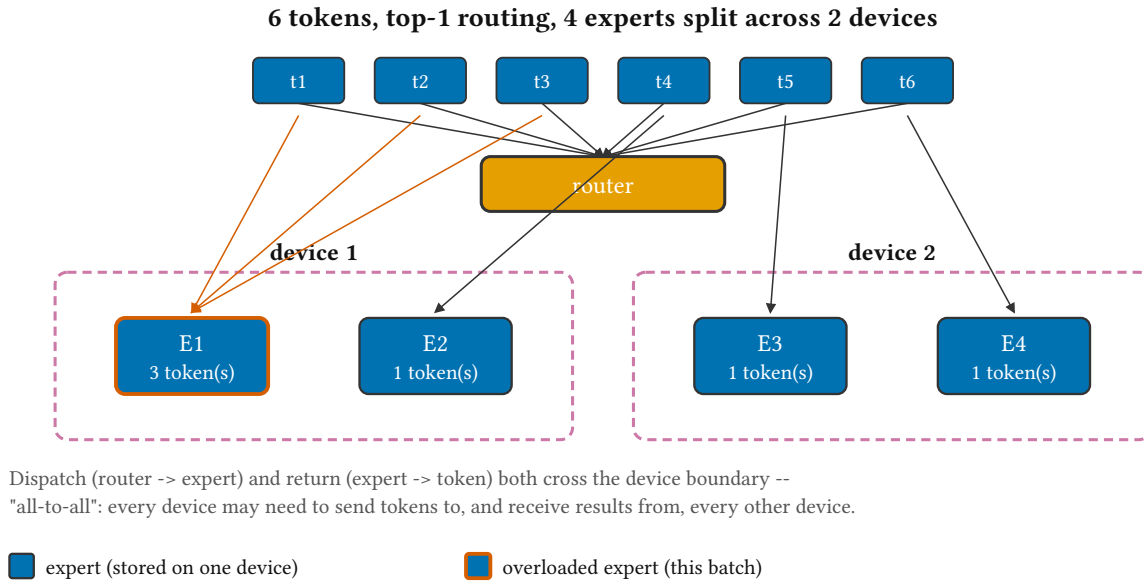


Figure 17: Tokens dispatch to whichever device holds their selected expert; routing can be imbalanced (E1 here gets 3 of 6 tokens), and results must return to their originating token.

7.5 Worked example

Parameters. $E = 8$, $k = 2$, $p_e = p_s = 100$ M, $p_{\text{base}} = 1.2$ B (verified programmatically; see the Build and Version Note):

Table 11: $E/k = 4$, but the complete model’s ratio is $1.4 - p_{\text{base}}$ and p_s stay active either way

Quantity	Total	Active	Ratio
Routed expert bank	800M	200M	4.0 ($= E/k$)
Complete model	2.1B	1.5B	1.4

Routing imbalance. 6 tokens, top-1, 4 experts (matching Figure 17): assignment $\{E_1 : 3, E_2 : 1, E_3 : 1, E_4 : 1\}$ against a nominal balanced load of 1.5/expert. At capacity factor 1.2 (capacity = 2 per expert, verified programmatically), E_1 overflows by 1 token. Neither this ratio nor this overflow count is a universal law – both depend on the specific config and routing outcome, not stated here as one.

7.6 Compare and contrast

Table 12: MoE does not automatically mean lower latency, lower memory, better quality, or balanced routing – each row is a tradeoff

	Dense FFN	Routed MoE	MoE + shared expert
Expert paths/token	1 (always)	k of E	k of E , plus 1 shared
Total parameters	smallest	largest	largest
Active parameters	equals total	$\ll$ total (bank-level)	$\ll$ total (bank-level)
Routing required?	no	yes	yes
Load-balancing concern	none	yes	yes
Distributed communication	none extra	all-to-all (if expert-parallel)	all-to-all (if expert-parallel)
Principal benefit	simplicity	more capacity, compute roughly held	also reduces routed redundancy
Principal complication	none	routing, imbalance, communication	also costs always-active compute

7.7 Check your understanding

Full solutions are in the answer key (Section 14):

1. **Explain.** What distinguishes a dense feed-forward block from a sparse MoE block?
2. **Explain.** Why are active model parameters not the same as active *routed-expert-bank* parameters?
3. **Compare.** Is MoE sparsity the same axis as Chapter 4’s sparse attention?
4. **Misconception check.** True or false: a 64-expert top-2 MoE activates only 2/64 of the whole model.
5. **Misconception check.** True or false: inactive experts consume no memory.

6. **Calculation.** For $E = 16$, $k = 4$, $p_e = 50$ M, no shared expert, $p_{\text{base}} = 2$ B: compute the routed-bank and complete-model total/active parameters and ratios.
-

(Space intentionally left for written answers if you are using this as a workbook exercise, not only a reading.)

7.8 Applied exercise

A serving team must choose between a dense model and a MoE model with the same *active* parameter count. Using this chapter’s total/active distinction (not benchmark numbers), predict which has the larger serving memory footprint, and state the one architectural fact that makes your answer certain regardless of which specific models are compared.

7.9 Interview lens

“A model has 64 experts and activates 2 per token. Explain what that tells you – and what it does not tell you – about parameters, FLOPs, memory, and communication.” A strong answer states that it tells you the *routed-expert-bank*’s activation ratio (2/64), but not the whole model’s total/active ratio (base and shared-expert parameters are unaffected), not exact FLOPs (routing and combination cost something too), not memory footprint (all 64 experts must still be stored), and not communication cost (which depends on how routing lands across devices, not on k or E alone).

7.10 Chapter recap

- MoE replaces the feed-forward sublayer with a router, top- k selected experts, and a weighted combination – attention is unchanged.
- E/k is the routed-expert-bank’s own total/active ratio, not the complete model’s – base and shared-expert parameters stay active regardless of routing.
- Inactive experts are still stored; sparsity changes compute, not memory footprint.
- Load-balancing losses encourage, but do not guarantee, balanced routing; capacity factors and overflow handling are implementation-dependent.
- Shared experts trade always-active compute for reduced redundancy across routed experts – not a universal requirement.
- Expert parallelism’s all-to-all communication is an unavoidable architectural consequence once experts are distributed, and can offset the sparse-compute benefit.
- Model capacity, computation, memory footprint, and communication are four separate quantities – no single number answers all four.

7.11 Sources and further reading

Primary sources: [13] (sparsely-gated MoE and its top- k routing and load-balancing losses), [14] (top-2 gating, its auxiliary load-balancing loss, and expert-parallel sharding), [15] (top-1 routing and a different load-balancing loss form – these formulations are genuinely not equivalent, not restated here as one “the” loss), [12] (DeepSeekMoE’s shared-expert motivation and compute tradeoff, verified against the paper’s full text, not a snippet). Secondary/supporting: [2]. Detailed distributed-training implementation, collective-communication algorithms, kernel optimization, inference-engine configuration, expert quantization, model-specific benchmark comparisons, post-training, and a survey of routing research are all out of scope for this chapter.

8 Chapter 6: Reading Modern LLM Architectures: Three Designs, Three Sets of Tradeoffs

A Ten-Question Checklist, Applied to Three Real Model Families

8.1 Learning objectives

By the end of this chapter you should be able to:

- Apply a compact, repeatable checklist to read a real model’s technical report or config and identify its block structure, normalization, position representation, sequence-mixing mechanism, head counts, feed-forward type, and (if MoE) expert counts.
- Compare one dense, one sparse-MoE-with-compressed-attention, and one hybrid-attention model on architecture alone, not on leaderboard scores.
- State what cannot be inferred about a model’s quality, latency, or cost from its architecture alone.

8.2 Why this matters

Chapters 2 through 5 taught individual mechanisms one at a time. Real models combine a small number of these mechanisms in different ways, and the skill this chapter teaches is reading which ones, from a model’s own documentation, using a fixed checklist rather than an ad hoc scan. Model families change quickly; the three case studies below are each pinned to an exact, dated version, not stated as a permanent fact about “the best” or “the latest” model – the checklist itself does not go stale the same way.

8.3 The architecture-reading checklist

1. What is the block structure – how are layers/blocks arranged and repeated?
2. Where is normalization applied?
3. What represents position?
4. What is the attention or sequence-mixing mechanism?
5. How many query and key/value heads (or equivalent) are there?
6. Is the feed-forward path dense or MoE?
7. If MoE: what are the total and active expert counts?
8. What state grows with context length?
9. Which components are active for every token, regardless of routing?
10. What training or serving complication follows from these choices?

The three case studies below answer these questions concretely; the architecture ledger (Table 13) collects every answer in one place.

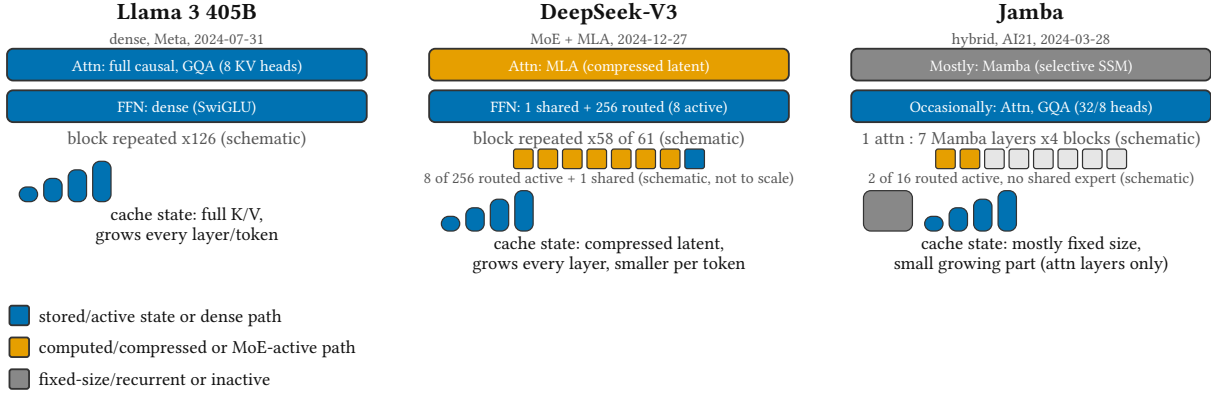


Figure 18: Three case-study architectures redrawn in one visual grammar. Repeated blocks and layer counts are schematic – see the ledger for exact counts. The one distinction worth noticing: each panel’s cache-state icon at the bottom (growing bars vs. a fixed box).

8.4 Three case studies

Llama 3 405B: a dense baseline. Meta’s Llama 3 405B ([16], 2024-07-31) is a 126-layer dense decoder: GQA ($H_q = 128$, $H_{kv} = 8$, group size 16, ch. 3) and RoPE ($\theta = 500,000$) in every layer, paired with a dense SwiGLU feed-forward block (ch. 2) – no routing, no MoE, no attention-mechanism variation across layers. This has the effect of keeping compute and memory uniform across positions and layers, at the cost of a KV-cache that grows every layer, every token (ch. 3), and of all 405B parameters being active for every token (ch. 5’s total/active distinction collapses to one number here). Do not infer from this uniformity that Llama 3 is cheaper or simpler to *train* than the other two – architecture alone does not settle that.

DeepSeek-V3: sparse experts, compressed attention. DeepSeek-V3 ([17], 2024-12-27, rev. 2025-02-18) combines two *independent* sparsity mechanisms: Multi-head Latent Attention (MLA, ch. 4) compresses each token’s key/value information into one shared latent ($d_c = 512$ plus a $d_{\text{rope}} = 64$ decoupled-RoPE component, per [11]) instead of per-head K/V, and a DeepSeekMoE feed-forward path (ch. 5) routes each token to 8 of 256 experts plus 1 always-active shared expert in all but its first 3 layers. This has the effect of holding active compute (37B) far below total capacity (671B) while shrinking the per-token cache footprint versus full multi-head attention – but it adds an all-to-all step wherever experts are split across devices (ch. 5), and reconstructing attention from the compressed latent adds computation MLA alone does not remove. Do not infer that MLA and MoE are “the same kind of sparsity” from their co-occurrence – one compresses attention state, the other narrows which feed-forward parameters run; they are independent mechanisms.

Jamba: attention interleaved with a recurrent mechanism. Jamba ([18], 2024-03-28) interleaves 4 attention layers (GQA, $H_q = 32$, $H_{kv} = 8$, group size 4) with 28 Mamba layers – a selective state-space mechanism ([19]) – across 4 repeated blocks (a 1:7 attention:Mamba ratio), plus a 16-expert, top-2 MoE path every other layer. This has the effect of replacing most of the growing KV-cache with a fixed-size recurrent state: Jamba’s own comparison places its cache at roughly 4GB at a 256K-token context, against 32-128GB for the fully-attention baselines it compares against ([18]) – while keeping a small attention-based component for exact, long-range lookup a fixed-size state cannot easily provide. Do not infer that a smaller *reported* cache number means uniformly lower cost across every workload: the comparison is Jamba’s own, at one stated configuration, not an independent benchmark.

8.5 Architecture ledger

Table 13: Architecture ledger – every cell traceable to the primary source cited in the text; “not applicable” is used rather than a guessed number

Attribute	Llama 3 405B	DeepSeek-V3	Jamba
Version / date	2024-07-31	2024-12-27 (rev. 2025-02-18)	2024-03-28
Total / active params	405B / 405B	671B / 37B	52B / 12B
Depth / hidden dim	126 layers / 16,384	61 layers / 7,168	32 layers (4x8) / 4,096
Normalization	RMSNorm (per Llama 2)	RMSNorm, plus post-latent norm	RMSNorm
Position representation	RoPE ($\theta = 500,000$)	RoPE, decoupled (MLA)	none (implicit, via Mamba)
Heads / cache representation	$H_q = 128$, $H_{kv} = 8$ (GQA-16)	$n_h = 128$, $d_c = 512 + d_{\text{rope}} = 64$	$H_q = 32$, $H_{kv} = 8$ (GQA-4, attention layers only)
Sequence-mixing pattern	uniform full attention	uniform MLA-compressed attention	hybrid: attention 1/8, Mamba 7/8
Feed-forward type	dense (SwiGLU)	dense (3 layers), then MoE	dense, plus MoE every other layer
Expert counts, if MoE	not applicable	1 shared + 256 routed, 8 active	16 routed, 2 active, no shared
Documented context length	128K	128K (via YaRN)	256K
Cache-relevant state	full K/V, grows every layer	compressed latent, grows every layer	mostly fixed size, small growing part
Principal tradeoff	uniform cost, full cache growth	large capacity, compressed cache, routing/reconstruction overhead	fixed-size state, smaller-scale MoE comms, newer mixing mechanism

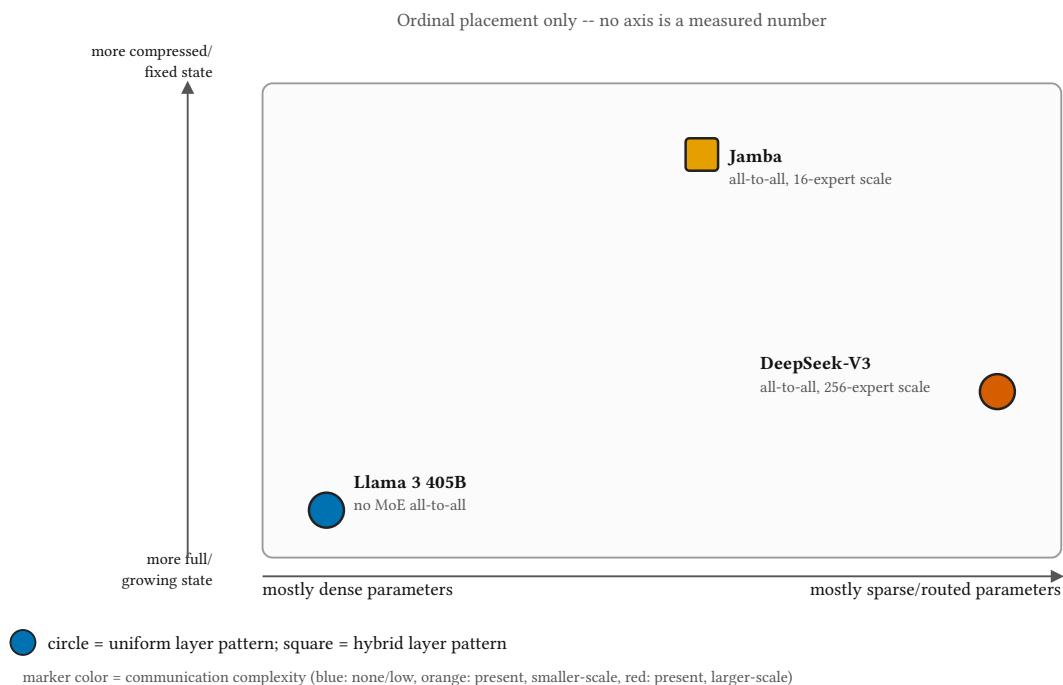


Figure 19: A qualitative tradeoff map. Positions are ordinal, not measured – there is no unit on either axis, and no coordinate here is a benchmark result. Marker shape encodes uniform vs. hybrid layer pattern; marker color encodes communication complexity at three qualitative levels, each traceable to a documented expert count.

Common misconception

“A more sophisticated architecture is necessarily a better model.” It is not automatically so. Measured model quality depends on architecture together with training data, training recipe, optimization, evaluation methodology, and serving implementation – none of the three case studies’ architectural choices alone determines how well, how cheaply, or how reliably the resulting model performs.

8.6 Check your understanding

Full solutions are in the answer key (Section 15):

1. **Quick check.** Given $H_q = 128$ and $H_{kv} = 8$ (Llama 3’s configuration), what GQA group size does this imply, and is this MHA, MQA, or GQA? _____
2. **Quick check.** DeepSeek-V3 routes each token to 8 of 256 experts plus 1 shared expert. Does the 8-of-256 ratio describe the whole model’s active-parameter fraction, or only the routed expert bank’s? _____
3. **Compare.** Llama 3 and Jamba both use GQA in their attention layers. Which of the two carries a per-token state that keeps growing with context length in *every* layer, and which does not? _____
4. **Misconception check.** True or false: a more sophisticated architecture is necessarily a better model. _____

(Space intentionally left for written answers if you are using this as a workbook exercise, not only a reading.)

8.7 Applied exercise

A 4th, unnamed model’s configuration – not one of the three case studies above – states: 32 query heads, 8 KV heads, alternating local/global attention, 64 routed experts with top-2 routing plus 1 always-active shared expert, and recurrent layers interleaved between attention layers. Using only this configuration and this chapter’s checklist: (a) state the GQA group size; (b) identify which component’s state grows with context length and which does not; (c) state the active-vs-total expert distinction; (d) name one likely communication concern; (e) name which chapters 2-5 mechanisms are present; and (f) state at least one conclusion that cannot be drawn from the config alone.

8.8 Interview lens

“Given a model configuration and technical report, walk me through how you would predict its memory, compute, and communication bottlenecks without overclaiming benchmark performance.” A strong answer works through this chapter’s checklist item by item: head counts and cache representation for memory; dense-vs-MoE and active parameter count for compute; whether experts or layers are split across devices for communication – and explicitly declines to predict measured latency, throughput, or quality from these facts alone, naming training data, training recipe, and serving implementation as the pieces architecture cannot supply.

8.9 Chapter recap

- A fixed, 10-question checklist lets you read block structure, normalization, position representation, sequence-mixing mechanism, head counts, feed-forward type, and expert counts off any model’s own documentation.
- Llama 3 405B is a uniform dense baseline: every parameter active for every token, a KV-cache that grows in every layer.
- DeepSeek-V3 combines two *independent* sparsity mechanisms – MLA’s per-token attention compression and DeepSeekMoE’s routed feed-forward sparsity – not one mechanism wearing two names.
- Jamba replaces most of its attention layers with a fixed-size recurrent (Mamba) state, keeping only a small fraction of layers as attention for exact long-range lookup.
- Architecture facts are comparatively stable and traceable to a technical report; benchmark scores are separately reported and more time-sensitive – this chapter deliberately stays on the former.
- A more sophisticated architecture is not necessarily a better model: data, training recipe, optimization, evaluation, and serving implementation all matter too.

8.10 Sources and further reading

Primary sources: [16] (Llama 3 405B’s architecture, verified directly against the paper’s Table 3 and Section 3.2.1, not a search snippet), [17] (DeepSeek-V3’s architecture, verified directly against Section 4.2’s hyper-parameters), [18] (Jamba’s architecture and its own KV-cache comparison table, verified against the paper’s full text plus its published config), [19] (Gu and Dao’s selective state-space mechanism, the recurrent update Jamba’s non-attention layers use – resolving a sourcing gap left open in ch. 4). Secondary/supporting: [2], [20], [11] (MLA’s compression mechanism, taught in ch. 4). A model-specific benchmark comparison, a survey of every current model, and a release-history timeline are all out of scope for this chapter; undocumented proprietary architectures are not discussed.

9 Chapter 7: From Architecture to Systems Behavior: Memory, Compute, Communication, and Throughput

Five Resources, Two Operating Modes, One Refusal to Guess

9.1 Learning objectives

By the end of this chapter you should be able to:

- Separate weight memory, KV-cache/state memory, activation memory, temporary workspace, optimizer state, and communication as distinct resource categories, and explain why training and inference have different memory profiles.
- Explain why prefill and decode stress hardware differently, and distinguish latency, throughput, and goodput.
- Trace the systems consequences (memory, compute, communication, implementation complexity) of GQA/MQA, MLA, sliding-window attention, MoE, and hybrid recurrent layers.
- State what cannot be inferred about wall-clock performance from theoretical FLOPs or an architecture diagram alone.

9.2 Why this matters

Chapters 2 through 6 taught what an architecture *is*. This chapter asks what that architecture *costs* to run – and insists on keeping five different resources (weights, cache/state, activations, communication, and temporary workspace) separate, since collapsing them into one number is the most common way a memory or performance estimate goes wrong. This chapter synthesizes consequences already implied by chapters 2-6; it is not a training-systems or inference-serving textbook.

9.3 The resource ledger

No single formula covers every resource a model touches. Weights persist across both training and inference. Activations, gradients, and optimizer state exist only during training. The KV-cache or recurrent state (ch. 3, ch. 6) is request-specific: it grows and shrinks with which sequences are currently being served, not with the architecture alone. Temporary kernel workspace is implementation-dependent – it varies by serving engine, not by architecture.

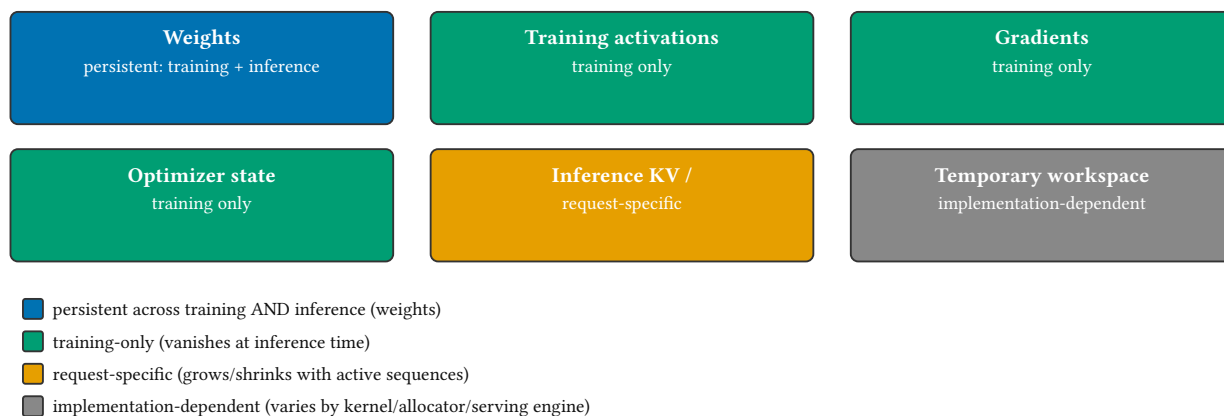


Figure 20: Six resources, four lifecycles: persistent across training and inference (weights); training-only (activations, gradients, optimizer state); request-specific (inference KV/recurrent state); and implementation-dependent (temporary workspace).

Table 14: A logical estimate and its measured counterpart are never the same number – allocator overhead, fragmentation, and framework buffers separate them

Resource	Logical (architecture) quantity	Measured (runtime) quantity
Weights	parameter count $\times$ bytes/elem (ch. 2, ch. 5)	allocated model memory (profiler)
KV-cache / recurrent state	ch. 3/ch. 6 formula	profiler reading, includes allocator overhead
Communication	bytes/step implied by a parallelism strategy	measured network/interconnect traffic

9.4 Prefill versus decode

Prefill processes every prompt position at once and can expose substantial parallel matrix work – it is often compute-heavy. Decode processes one new position at a time, repeatedly reading the retained KV-cache or recurrent state – it is often memory-bandwidth-sensitive, since a single new token’s FLOPs are small relative to the bytes moved rereading that state. Neither claim is universal: larger batches raise arithmetic intensity and can shift decode back toward compute-bound, but they also multiply cache state, and long contexts increase cache traffic regardless of batch size. Architecture and serving engine interact – this chapter states tendencies, not laws.

9.5 Architecture-to-systems consequence map

Every mechanism taught so far trades one resource for a complication elsewhere – none improves every column below.

Qualitative markers only -- no cell is a measurement

Mechanism	Weight memory	Cache/state memory	Compute	Communication	Impl. complexity
GQA / MQA	same	lower	~same	none new	low
MLA	similar	much lower	higher	none new	higher
Sliding / local attn.	same	bounded	lower	none new	moderate
MoE	much higher	same	lower (active)	new (all-to-all)	higher
Hybrid recurrent layers	same/lower	much lower	~comparable	none new	higher
Looped depth (preview -- ch. 8)	lower	not auto-reduced	NOT reduced	none new	preview only

Looped depth’s row previews only -- see chapter 8 for the full mechanism.

MTP/speculative decoding are cross-referenced in the text, not shown as a row here (not a main architecture change).

Figure 21: No mechanism improves every resource dimension. Qualitative markers only – no cell is a measurement. The looped-depth row is a systems-oriented preview: weight sharing across repeated block applications lowers distinct parameter count, but does not automatically lower compute (the repeated applications still run) or cache/state size – full explanation deferred to ch. 8.

An auxiliary multi-token-prediction module (a training-time design, [21]) may itself become an inference-time draft mechanism for speculative decoding – a serving technique this workbook does not teach; both are covered in depth in a future inference workbook, per this series’ own cross-workbook topic roadmap.

9.6 Parallelism, briefly

Table 15: Each type addresses one resource pressure and introduces its own communication shape – not derived further here; see [8] and a future training workbook

Type	Resource pressure addressed	Communication introduced
Data	replicates computation across data partitions	gradient synchronization
Tensor	splits large matrix ops across devices	activation/gradient all-reduce per layer
Pipeline	assigns layer ranges to different devices	activations forwarded between stages
Expert	distributes routed experts (ch. 5)	token dispatch/combine (all-to-all)
Sequence/context	distributes sequence-length computation/state	state exchanged across sequence shards

Common misconception

“The architecture with fewer theoretical FLOPs must be faster.” It need not be. Wall-clock performance also depends on arithmetic intensity, memory traffic, communication, batching, kernel implementation, scheduling, and hardware utilization – any one of which can dominate over raw FLOPs.

9.7 Check your understanding

Full solutions are in the answer key (Section 16):

1. **Quick check.** Which resource-ledger category does the KV-cache belong to, and which category does an optimizer’s momentum/variance state belong to?

2. **Quick check.** Is a temporary attention-kernel workspace training-only, inference-persistent, request-specific, or implementation-dependent? _____
3. **Compare.** Contrast one prefill step and one decode step on the same request, using the resource ledger and arithmetic intensity. _____
4. **Misconception check.** True or false: the architecture with fewer theoretical FLOPs must be faster.

(Space intentionally left for written answers if you are using this as a workbook exercise, not only a reading.)

9.8 Applied exercise

Model A: dense, GQA ($H_q = 32$, $H_{kv} = 8$, 32 layers), 7B parameters, long retained context, high concurrency. Model B: MoE + MLA (64 routed experts, 8 active, $p_{\text{base}} = 2$ B; $d_c = 512$, $d_{\text{tope}} = 64$), same context

length and concurrency as Model A, expert weights distributed across devices. Using only ch. 3's and ch. 5's own formulas: (a) which model likely uses less logical cache; (b) which has more total weight storage; (c) which introduces expert communication; and (d) why latency cannot be concluded from any of this without benchmarking.

9.9 Interview lens

“Given an architecture report, how would you estimate likely serving bottlenecks, and which conclusions would you refuse to make without benchmarking?” A strong answer separates the five resource-ledger categories, applies the architecture-to-systems consequence map mechanism by mechanism, and explicitly refuses to predict measured latency, throughput, or ranking from architecture facts alone.

9.10 Chapter recap

- Five resource categories – weight memory, cache/state memory, activation memory, temporary workspace, and communication – answer different questions; no single formula covers all of them.
- Logical/architectural estimates and measured runtime numbers are never the same number.
- Prefill is often compute-heavy; decode is often memory-bandwidth-sensitive – both are tendencies, not universal laws, and batching/context length/serving engine all interact with them.
- Every mechanism trades one resource dimension for a complication elsewhere; looped depth's weight-sharing benefit does not automatically reduce compute or cache/state (full treatment: ch. 8).
- An MTP module may become a speculative-decoding draft source, but MTP and speculative decoding are not the same thing, and neither is taught here.
- Theoretical FLOPs are necessary but not sufficient to predict wall-clock performance.

9.11 Sources and further reading

Primary sources: [8] (KV-cache math, attention FLOPs, and roofline/hardware-aware cost analysis – the core narrative source for this chapter's resource ledger and prefill/decode framing), [22], [23] (inference-serving systems companions for the same distinctions). Secondary/supporting: [24] (parallelism strategies, access partial – cited only for the conceptual map, not a specific communication-volume claim), [25], [26], [27] (the looped-depth preview row's sources, per this series' own cross-workbook topic roadmap), [21], [28] (the MTP cross-reference). A full derivation of communication-volume formulas, a serving-engine implementation guide, and the speculative-decoding algorithm itself are all out of scope for this chapter.

10 Answer Key: Chapter 1

Full solutions to Chapter 1’s “Check your understanding” questions (Section 3).

1. Quick check. What does the residual stream carry from one block to the next?

Answer. The running sum of the original embedding plus every sublayer update added so far – not just the most recent sublayer’s raw output. Per the residual update $x^{(l+1)} = x^{(l)} + f(\text{Norm}(x^{(l)}))$, each block adds its sublayer’s output onto the existing stream value; it never replaces the stream with just the new sublayer’s output.

Common trap: “the output of the most recent attention or MLP sublayer” – that describes only the update being added in that step, not the accumulated stream value.

2. Explain. Why does a decoder-only objective allow one-token- at-a-time generation?

Answer. Because the causal mask restricts every position to attending only to itself and earlier positions, the same forward-pass structure used in training also directly defines how to generate: at each position, produce a next-token distribution conditioned only on what came before. A bidirectional encoder’s positions can attend to *later* positions too, so its representation at a given position already depends on tokens that would not exist yet during generation – that structure cannot be reused one token at a time the way a causal one can.

Common trap: “decoder-only models are just smaller/simpler” – size is unrelated; the difference is specifically about which positions are allowed to attend to which.

3. Calculation. $d_{\text{model}} = 8$, $H_q = 2$, $S = 5$: find d_{head} and the shape of $QK^\top$ for one head.

Answer. $d_{\text{head}} = d_{\text{model}}/H_q = 8/2 = 4$. The raw attention score matrix for one head is $(S, S) = (5, 5)$ – one score per query/key pair, regardless of d_{head} .

Common trap: $(S, d_{\text{head}}) = (5, 4)$ – this confuses the per-head attention *output* shape (which is (S, d_{head})) with the score matrix shape (always (S, S)).

4. Misconception check. “Causal masking means each token can only attend to the immediately previous token.”

Answer. False. Causal masking allows a position to attend to *every* position at or before it ($j \leq i$), not only the single immediately preceding position. The mask is 0 (allowed) for every $j \leq i$ and $-\infty$ (blocked) only for $j > i$ – a full lower-triangular region, not a single diagonal band.

Why: conflating causal masking with a fixed local window (sliding-window attention, a separate mechanism covered in a later chapter). Causal masking alone imposes no limit on how far back a position can look.

5. Misconception check. “The residual stream is a separate memory database the model writes facts into and reads back later.”

Answer. False. The residual stream is a single vector per position, updated additively in place – not a separate, queryable store. Every sublayer’s output is added directly onto the same fixed-width vector that already exists at that position; there is no separate storage structure and no mechanism for a later block to selectively “query” what an earlier block specifically wrote. It can only read the accumulated sum.

Why: taking this chapter’s own receipt analogy too literally – a real receipt’s lines stay individually readable; the residual stream’s contributions are summed into one vector and cannot be individually read back out. This is exactly the analogy’s stated breakdown point.

6. Design exercise (applied exercise). Full tensor-shape trace for your own toy config.

Answer. For config $(d_{\text{model}}, L, H_q)$ with d_{model} divisible by H_q : input vector $(d_{\text{model}},)$; $Q/K/V$ before splitting, $(d_{\text{model}},)$ each (or (S, d_{model}) for a full sequence); after splitting into heads, (H_q, d_{head}) where $d_{\text{head}} = d_{\text{model}}/H_q$; per-head attention output, $(d_{\text{head}},)$ per head; concatenated across heads, back to $(d_{\text{model}},)$; final vector added to the residual stream, $(d_{\text{model}},)$ – matching the input shape exactly, since addition requires matching shapes.

Common trap: forgetting to concatenate the per-head outputs back to d_{model} width before the output projection and residual addition – leaving the answer at the per-head shape.

7. Interview practice. “Walk me through what happens differently between processing a 500-token prompt versus generating the 501st token.”

Answer. A strong answer: (1) names prefill – conceptually, all 500 positions evaluated together, building per-layer K/V state for every position (a serving engine may chunk this for scheduling reasons without changing the logical picture); (2) names the decode step – exactly one new position processed, computing only its own new K/V and reusing the 500 cached K/V rows unchanged; (3) explicitly states that generating the 501st token does not reprocess or re-read the first 500 tokens.

Interview criteria: does the answer name both phases correctly, does it state what specifically is reused (K/V rows) versus recomputed (the one new position’s K/V), and does it avoid the “re-reads the whole prompt” framing this chapter’s headline misconception warns against?

Common trap: “decoding is faster because the prompt is already processed,” without stating what specifically is reused or why that reuse is what makes it faster.

11 Answer Key: Chapter 2

Full solutions to Chapter 2’s “Check your understanding” questions (Section 4).

1. Calculation. RoPE, positions (1,0) and (3,2), both relative offset 1: what’s guaranteed, what isn’t?

Answer. RoPE guarantees the *angle* between the rotated query and key vectors is identical in both cases, because each vector is rotated by an amount tied to its own position, so only the relative offset (1, in both cases) survives. It does **not** guarantee the dot products are numerically equal in general – the dot product also depends on the original (pre-rotation) vectors’ magnitudes and directions, which can differ between the two pairs.

Common trap: claiming RoPE guarantees identical dot products for any two pairs with the same relative offset – this conflates the angle (guaranteed to match) with the dot product (which also depends on vector magnitudes).

2. Compare. Two operations RMSNorm omits relative to LayerNorm.

Answer. Mean-centering (subtracting the vector’s mean before rescaling) and the learned bias term (an additive per-dimension shift after rescaling). LayerNorm computes $(x - \text{mean}(x))/\text{std}(x) \odot \gamma + \beta$;

RMSNorm computes $x/\text{RMS}_\epsilon(x) \odot \gamma$ – no mean subtraction and no β term (the ϵ inside RMS_ϵ is a fixed numerical-stability constant, not a learned bias).

Common trap: naming only the missing bias and missing that mean-centering is also gone – dropping mean-centering is the more consequential of the two omissions.

3. Explain. Why three matrices for a gated feed-forward layer?

Answer. The third matrix (W_3 , the “value” branch) provides a second, purely linear projection of the input that gets elementwise-multiplied against the first matrix’s activated output (W_1 , the “gate” branch) before the shared output projection (W_2). W_1 alone only produces one activated signal; W_3 gives something for that signal to gate.

Common trap: describing the third matrix as making the network “deeper” – it does not; W_1 and W_3 are parallel branches computed from the same input, not a longer sequential chain.

4. Calculation. Plain MLP, $d_{\text{model}} = 16$, $d_{\text{ff}} = 64$: parameter count.

Answer. $2 d_{\text{model}} d_{\text{ff}} = 2 \times 16 \times 64 = 2048$ parameters, from this chapter’s plain-MLP formula instantiated at these dimensions and checked programmatically (see the Build and Version Note).

Common trap: computing only one matrix’s parameter count (1024) and forgetting the second, or applying the gated-MLP (3x) formula by mistake.

5. Misconception check. Pre-norm’s popularity: stability or quality?

Answer. Mainly training stability at depth, not final model quality. Pre-norm trades away some representational capacity for a more stable gradient path through an un-normalized residual stream as L grows. Sandwich-norm’s existence (paying for a *second* normalization step around the same sublayer) is direct evidence that pre-norm alone is understood to give something up, not that it is a strict improvement.

Common trap: treating pre-norm as simply “the new correct default” with no acknowledged tradeoff.

6. Misconception check. “RoPE adds a learned position vector to every token embedding.”

Answer. False. RoPE rotates the query and key vectors *inside* the attention computation, at every layer that has attention – it adds nothing to the embedding table, and the rotation angles are fixed by position, not learned.

Why: conflating RoPE with learned absolute positional embeddings (an older, different scheme that *does* add a learned per-position vector directly to the embedding).

7. Misconception check. “SwiGLU makes the feed-forward block sparse like an MoE layer.”

Answer. False. Every weight in a SwiGLU feed-forward layer is used for every token, exactly like a plain feed-forward layer. Sparsity, in the mixture-of-experts sense, means a *router* activates only a subset of available parameters per token – SwiGLU has no router and always uses 100% of its own weights.

Why: the word “gated” sounds like it could mean “selecting a subset,” but the gate here is a continuous, dense elementwise multiply applied to every token, not a discrete routing decision.

8. Design exercise (applied exercise). Reading a model config excerpt.

Answer. Given `hidden_size: 2048, num_attention_heads: 16, intermediate_size: 5504, hidden_act: "silu", rms_norm_eps: 1e-5, tie_word_embeddings: false, rope_theta: 10000.0:`

- **Normalization:** RMSNorm – the `rms_norm_eps` field name is the giveaway.
- **Gated feed-forward:** yes – `hidden_act: "silu"` matches SwiGLU’s Swish/SiLU convention, and `rope_theta` separately confirms RoPE is in use.
- **Ratio check:** $5504/2048 \approx 2.69$, close to $\frac{2}{3} \times 4 \approx 2.67$ – a *gated-compensated* ratio, not the plain 4x ratio.
- **Embeddings:** untied (`tie_word_embeddings: false` is explicit).

Common trap: assuming any roughly-4x `intermediate_size` implies a plain (ungated) MLP – the compensated gated ratio is close enough to other plausible ratios that the activation-function field is a more reliable signal than the ratio alone.

9. Interview practice. “Why RoPE instead of learned positional embeddings?”

Answer. A strong answer states RoPE’s relative-position mechanism as the reason, and names at least one concrete consequence: no learned positional parameters at all, and generally better behavior beyond the training length than a learned table with a fixed maximum.

Interview criteria: does the answer name the actual mechanism rather than asserting RoPE is “more modern”; does it name a specific, checkable consequence rather than a vague quality claim?

10. Interview practice. “Walk me through what SwiGLU changes about the MLP?”

Answer. A strong answer describes the extra parallel branch and the elementwise multiply before the shared output projection, and explicitly states the parameter tradeoff: at matched expansion width, three matrices cost 1.5x what two matrices cost, which is why real models often shrink the expansion width to compensate.

Interview criteria: does the answer proactively raise the parameter-cost tradeoff, rather than needing to be prompted for it – a systems-flavored interviewer will very likely ask a follow-up about cost if the candidate doesn’t raise it first.

12 Answer Key: Chapter 3

Full solutions to Chapter 3’s “Check your understanding” questions (Section 5).

1. Explain. What is the one structural difference between MHA, MQA, and GQA – and what stays identical across all three?

Answer. The one difference is H_{kv} : MHA sets $H_{kv} = H_q$ (every query head has its own key/value head), MQA sets $H_{kv} = 1$ (all query heads share one), and GQA sets $1 < H_{kv} < H_q$ (groups of query heads each share one). H_q itself, the query projection’s shape, the output projection’s shape, and the number of query-side attention computations are identical across all three.

Common trap: describing MHA/MQA/GQA as three unrelated mechanisms rather than three points on the same H_{kv} axis.

2. Calculation. For $H_q = 8$ query heads and $H_{kv} = 2$ key/value heads, what is the group size, and how many query heads share each key/value head?

Answer. Group size $= H_q / H_{kv} = 8 / 2 = 4$. Each key/value head is shared by 4 query heads.

Common trap: dividing H_{kv} by H_q instead of H_q by H_{kv} , which gives a fraction instead of a head count.

3. Calculation. Write the shapes of Q and K for $B = 1$, $T_q = 1$, $S = 100$, $H_q = 16$, $H_{kv} = 4$, $d_{\text{head}} = 64$.

Answer. $Q : (1, 1, 16, 64)$. $K : (1, 100, 4, 64)$.

Common trap: using S in place of T_q for Q 's second axis (or vice versa for K) – Q 's position axis is how many *new* positions this step computes queries for, while K/V 's position axis is how many positions are *retained* in the cache.

4. Quick check. Which two of P_Q , P_K , P_V , P_O change when H_{kv} changes, holding H_q and d_{head} fixed?

Answer. P_K and P_V . $P_Q = d_{\text{model}} H_q d_{\text{head}}$ and $P_O = H_q d_{\text{head}} d_{\text{model}}$ depend only on H_q (and $d_{\text{head}}/d_{\text{model}}$), not H_{kv} .

Common trap: assuming all four projections shrink together, since they are all part of “the attention block.”

5. Misconception check. “GQA reduces the number of query heads.”

Answer. False. H_q is unchanged by GQA (or MQA) – only H_{kv} changes; Figure 9 draws the same number of query heads in every panel. This misconception usually comes from conflating “fewer heads are involved in some sense” with “fewer query heads specifically,” when what actually shrinks is the key/value side only.

6. Misconception check. “If GQA cuts a model’s KV-cache memory by 4x, the whole model becomes 4x smaller.”

Answer. False. Table 7 shows the complete attention block (including the unchanged P_Q and P_O) shrinks by only 1.6x here, not 4x, and the feed-forward sublayers are untouched by H_{kv} entirely. The cache formula and the model’s total parameter count share one input (H_{kv}) but respond to it very differently – the cache formula has no $Q/O/MLP$ terms to hold anything back, while the parameter count does.

7. Calculation. For $L = 24$, $H_q = 16$ (MHA, so $H_{kv} = H_q = 16$), $d_{\text{head}} = 128$, $S = 4096$, $B = 1$, bf16: compute the logical KV-cache size in bytes and MiB.

Answer. $M_{KV} = 2 \times 1 \times 4096 \times 24 \times 16 \times 128 \times 2 = 805,306,368$ bytes = 768 MiB exactly (verified programmatically; see the Build and Version Note).

Common trap: forgetting the factor of 2 for keys and values, which would understate the answer by exactly half (384 MiB).

8. Design exercise (applied exercise). Compare a candidate MHA model and a candidate GQA model for a long-context, high-concurrency serving workload.

Answer. At the same base size and S , the GQA candidate’s per-sequence cache is smaller by the ratio $H_{kv, \text{GQA}}/H_q$ (per Equation 13), which at high concurrency directly increases how many sequences fit in a fixed memory budget – usually the dominant argument for GQA here. But the MHA candidate could still win on task-specific quality (if trained, not just uptrained, with full per-head capacity) or serving-stack maturity. A complete answer performs the ratio comparison *and* names a concrete non-cache factor, not a vague “it depends.”

9. Interview practice. “Walk me through why GQA reduces KV-cache cost, and tell me what it does not automatically make cheaper.”

Answer. A strong answer: (1) names H_{kv} , not H_q , as the quantity GQA changes; (2) states Equation 13’s linear dependence on H_{kv} as the mechanism; (3) proactively names what does *not* shrink proportionally – P_Q/P_O , the number of query-side attention computations, and the complete attention block’s total parameter count (per Table 7).

Interview criteria: does the answer name the specific mechanism rather than a vague “less memory” claim, and proactively name at least one thing that stays fixed size rather than needing to be prompted for it? Stopping at “GQA uses less memory” is the common trap here – a systems-flavored interviewer will very likely ask exactly that follow-up.

13 Answer Key: Chapter 4

Full solutions to Chapter 4’s “Check your understanding” questions (Section 6).

1. Quick check. *Answer.* The score matrix grows quadratically ($O(S^2)$); the KV-cache grows linearly in S (Equation 13). *Common trap:* assuming both scale the same way.

2. Explain. *Answer.* Rotating the compressed latent directly would couple the up-projection W^{UK} to a position-specific rotation matrix that can no longer be folded into W^Q , forcing every prior key to be recomputed each step. The decoupled rotary component carries positional information separately, preserving matrix absorption. *Common trap:* assuming positional info can simply be added to the latent like an ordinary embedding.

3. Explain. *Answer.* No. Reconstructing full K/V is always a *conceptually* valid derivation, but an optimized implementation may instead absorb W^{UK} into W^Q and W^{UV} into W^O , computing attention directly from the cached latent without ever materializing full K/V . *Common trap:* treating reconstruction as mandatory rather than one available (unabsorbed) path.

4. Misconception check. *Answer.* False, for a stacked model: one layer’s direct window is W , but stacking L windowed layers gives an *indirect* effective reach of $L \times W$ (Equation 14) – older information can still propagate through earlier layers. *Common trap:* confusing one layer’s direct window with the whole model’s effective reach.

5. Misconception check. *Answer.* False. Compressing the cache reduces stored *bytes*; end-to-end decode latency also depends on compute, kernel implementation, and whatever else is bottlenecking that step – a 4x cache cut is not a guaranteed 4x latency cut. *Common trap:* equating a memory ratio with a latency ratio (the same error Chapter 3 flagged for GQA).

6. Calculation. *Answer.* MLA elements/token = $(d_c + d_{\text{rope}})L = (512 + 64) \times 24 = 13,824$. GQA at $H_{kv} = 4$: $2H_{kv}d_{\text{head}}L = 2 \times 4 \times 128 \times 24 = 24,576$. Ratio = 0.5625 – MLA is about 1.8x smaller here (verified programmatically; see the Build and Version Note). *Common trap:* forgetting GQA’s factor of 2 (keys and values), which MLA’s single joint latent does not have.

7. Design exercise (applied exercise). *Answer.* A pure sliding window only ever gives *indirect* access ($L \times W$) to old tokens, which can fail to reach a token far outside that reach at high S . Combining a sliding window with occasional global layers/tokens restores direct access to specific old positions; adding GQA or MLA then reduces the memory cost of retaining that longer effective history at high concurrency. *Interview criteria:* does the answer name the specific failure mode (indirect-only reach, not “windows are bad”), and combine mechanisms rather than picking just one?

8. Interview practice. *Answer.* GQA primarily reduces KV-cache memory/bandwidth by sharing K/V heads, at some representational cost. Sliding-window attention reduces both compute and memory by restricting connectivity, at the cost of only indirect long-range access. MLA reduces memory/bandwidth by compressing what is stored per token, at the cost of implementation complexity (a separately-cached rotary component, optional matrix absorption). *Common trap:* collapsing any two of the three into “they all just save memory” – compute and access-pattern effects differ.

14 Answer Key: Chapter 5

Full solutions to Chapter 5’s “Check your understanding” questions (Section 7).

1. Explain. *Answer.* Dense: every token uses the same feed-forward parameters. MoE: a router selects top- k of E experts per token; only the selected experts run.

2. Explain. *Answer.* Active model parameters include p_{base} (and p_s , if any) which stay active regardless of routing; active *bank* parameters (kp_e) count only the routed experts.

3. Compare. *Answer.* No. MoE sparsity is *which parameters* activate (feed-forward); sparse attention (Ch. 4) is *which positions* are scored. Orthogonal axes.

4. Misconception check. *Answer.* False, unless restricted to the routed bank. p_{base}/p_s stay active, so the whole-model ratio is always closer to 1 than E/k .

5. Misconception check. *Answer.* False. Unselected experts are still stored weights that must fit in memory – sparsity affects compute, not storage. *Common trap:* equating “not activated” with “not stored.”

6. Calculation. *Answer.* Bank: total = $16 \times 50 \text{ M} = 800 \text{ M}$, active = $4 \times 50 \text{ M} = 200 \text{ M}$, ratio 4.0. Model: total = $2 \text{ B} + 800 \text{ M} = 2.8 \text{ B}$, active = $2 \text{ B} + 200 \text{ M} = 2.2 \text{ B}$, ratio ≈ 1.27 (verified programmatically).

7. Design exercise (applied exercise). *Answer.* The MoE model has the larger serving memory footprint. The certain fact: serving memory must hold (most of) *total* parameters regardless of which are active for any one token, and a MoE model’s total is by construction larger than its active count – while the dense model’s total equals its active count exactly. *Interview criteria:* does the answer name this structural fact rather than guessing from typical model sizes?

8. Interview practice. *Answer.* $2/64$ tells you the *routed bank*’s activation ratio only. It does not tell you: the whole model’s total/active ratio (base/shared parameters are unaffected); exact FLOPs (routing and combination cost something too); memory footprint (all 64 experts are still stored); or communication cost (depends on how routing actually lands across devices, not on k/E alone). *Interview criteria:* does the answer address all four (parameters, FLOPs, memory, communication) rather than stopping after one or two?

15 Answer Key: Chapter 6

Full solutions to Chapter 6’s “Check your understanding” questions (Section 8).

1. Quick check. *Answer.* Group size = $H_q/H_{kv} = 128/8 = 16$. Since $1 < H_{kv} < H_q$, this is GQA, not MHA ($H_{kv} = H_q$) or MQA ($H_{kv} = 1$).

2. Quick check. *Answer.* Only the routed expert bank’s ratio – attention, the shared expert, and every other always-active component also count toward the whole model’s active fraction, which stays closer to 1 than the bank-level k/E ratio (ch. 5).

3. Compare. *Answer.* Llama 3’s growing state is present in every one of its 126 layers (all attention). Jamba’s growing state exists only in its 4 attention layers out of 32 – the other 28 keep a fixed-size recurrent state instead.

4. Misconception check. *Answer.* False. Measured model quality depends on architecture together with training data, training recipe, optimization, evaluation methodology, and serving implementation – none of the three case studies’ architectural choices alone determines how well, how cheaply, or how reliably the resulting model performs. *Common trap:* treating one architectural feature (more experts, a newer mechanism) as a stand-in for overall quality.

5. Design exercise (applied exercise). *Answer.* (a) Group size = $32/8 = 4$. (b) Attention layers’ KV-cache grows with context; recurrent layers’ state stays fixed size. (c) 2 of 64 routed experts active per token plus 1 shared; all 64 still stored. (d) All-to-all dispatch if the 64 experts are device-split (ch. 5). (e) GQA (ch. 3), local/global attention (ch. 4), routed MoE with a shared expert (ch. 5), and a recurrent/state-space layer (ch. 6) are all present. (f) Quality, latency, training recipe, and serving cost cannot be read off the config alone. *Interview criteria:* separates the bank ratio from any whole-model claim; names a non-inferable conclusion.

6. Interview practice. *Answer.* Head counts/cache representation predict memory; dense-vs-MoE and active parameters predict compute; whether experts/layers are device-split predicts communication – latency, throughput, and quality are explicitly not predicted from architecture alone. *Interview criteria:* covers memory, compute, *and* communication; refuses to overclaim benchmark performance.

16 Answer Key: Chapter 7

Full solutions to Chapter 7’s “Check your understanding” questions (Section 9).

1. Quick check. *Answer.* The KV-cache is request-specific, persistent inference state – it grows with retained context/ concurrency, not architecture alone. An optimizer’s momentum/variance state is training-only – it does not exist at inference time.

2. Quick check. *Answer.* Implementation-dependent – a temporary attention-kernel workspace varies by serving engine/kernel, not architecture, unlike weight or cache memory.

3. Compare. *Answer.* Prefill processes every prompt position in parallel, often compute-heavy (high arithmetic intensity). Decode processes one new position at a time, repeatedly rereading retained state – often memory-bandwidth-sensitive (low arithmetic intensity). Neither is universal: larger batches raise intensity and can shift decode back toward compute-bound, but also multiply cache state.

4. Misconception check. *Answer.* False. Wall-clock performance also depends on arithmetic intensity, memory traffic, communication, batching, kernels, scheduling, and hardware utilization – any one can dominate over raw FLOPs. *Common trap:* treating fewer FLOPs as a guaranteed latency win.

5. Design exercise (applied exercise). *Answer.* (a) Model B uses less logical cache – MLA’s compressed latent is about 0.28x Model A’s GQA cache at the same context/concurrency (verified programmatically). (b) Model B has more total weight storage – its MoE total is about 1.2x Model A’s dense 7B, though Model B’s *active* parameters (~2.8B) are well below its own total. (c) Model B introduces expert commu-

nication (all-to-all, ch. 5); Model A introduces none. (d) Neither model’s latency follows from these logical quantities alone – batching, kernel implementation, hardware, and actual routing distribution still need benchmarking. *Interview criteria:* keeps cache, weight storage, and communication as three separate comparisons, not one “which model is better” verdict.

6. Interview practice. *Answer.* A strong walk-through separates the five resource-ledger categories, applies the consequence map mechanism by mechanism, and explicitly refuses to predict latency, throughput, or a quality ranking from architecture alone – naming benchmarking as the missing step. *Interview criteria:* names specific categories/mechanisms rather than “it depends”; refuses to overclaim.

17 Build and Version Note

- **Generated:** 2026-09-25
- **Git commit:** 2c48fb811f6792b26e8bc832704c3c317ff6c25e (pre-commit -- working tree had uncommitted changes at build time)
- **Source-registry verification date referenced by this build:** src-19..src-49 (primary/roadmap sources for Ch.1-7 and the series-wide topic roadmap) last verified 2026-09-22..2026-09-25; see sources/registry.yaml
- **Pilot status:** Seven-chapter pilot (Chapters 1-7 of 8) – Chapter 8 is planned but not drafted; Chapter 7 previews looped depth in one row only and cross-references (does not teach) multi-token prediction/speculative decoding (see outline.yaml’s ch7.drafting_note and config/series-topic-roadmap.yaml)

17.1 Provenance

- Tied-embeddings usage (Chapter 2) has no dedicated primary source; the sandwich-norm claim (Chapter 2) leans on a secondary source rather than a model’s own technical report. Both are tracked sourcing gaps, not unsupported claims.
- All worked-example numbers in Chapters 1-7 (attention weights, parameter counts, KV-cache sizes, receptive fields, MoE parameter/routing counts, ch. 6’s hypothetical-config GQA group size, ch. 7’s two-model resource diagnosis) are computed by version-controlled project scripts and checked by the project’s automated test suite, not hand-derived.
- Figures are generated from version-controlled scripts using a fixed visual-style configuration, not drawn freehand; each figure’s caption credits its scholarly/technical source where one applies.
- Chapter 3’s and Chapter 4’s KV-cache/latent-cache/attention-projection-parameter figures and formulas are LOGICAL, theoretical minimums, not measured GPU memory – they exclude allocator overhead, page metadata, fragmentation, and framework buffers. Chapter 7’s resource ledger makes this logical-vs-measured distinction explicit as its own table.
- Chapter 6’s three case-study models and the Mamba mechanism source were each verified by downloading the primary source PDF and extracting its full text (pdftotext), not by trusting a search snippet or a fetch tool’s own summary.
- This session also created config/series-topic-roadmap.yaml and reports/series_topic_roadmap.md, registering 12 new sources (src-38..src-49) for three cross-workbook topics (recurrent depth/looped transformers, multi-token prediction, speculative decoding) that Chapter 7 only previews or cross-references – their full treatment is assigned to future chapters/workbooks, not drafted this session. No official source confirms any specific proprietary model’s architecture (e.g. ‘GPT-6 Astra’); any such claim is treated as unconfirmed.

- Chapter 7 folds quantization/paging/continuous-batching into its resource-ledger framing (runtime vs. architecture) rather than keeping a standalone table; MTP and speculative decoding are each given only a short cross-reference, not a taught mechanism, per that chapter’s own scope fence.

This note is generated by `scripts/generate_build_note.py` at build time – do not hand-edit `includes/build-version-note.qmd`; regenerate it instead.

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